

Icons and Age-Based Accessibility

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(a.k.a. ACKs, as CSCI389 has taught me)

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List of Abbreviations

HCI	Human-Computer Interaction
AI	Artificial Intelligence
AIC	Akaike Information Criterion
ANOVA	Analysis of Variance
ARIA	Accessible Rich Internet Applications
DV	Dependent Variable
IO	Icon Only
IV	Independent Variable
IWTL	Icon with Text Labels
MTurk	Mechanical Turk
RT	Reaction Time
TO	Text Only
UI	User Interface
USD	United States Dollars
UX	User Experience
WCAG	Web Content Accessibility Guidelines
WWW	World Wide Web

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Abstract

In 2004, 70-98% of websites were found to be inaccessible [?], and one of the user demographics most negatively affected by inaccessible websites is older users [?]. Many Human-Computer Interaction researchers have looked at whether different variants of Internet icons create positive performance effects for older users, with generally inconclusive results.

The goal of this thesis is to measure whether different icon types (specifically text, icons, or icons with text labels) have a positive impact on older users's performance (as compared to younger users) in a web-based task-completion setting. Specifically, users were assigned to an icon type and then shown six unique websites each containing one task. This research was conducted online and asynchronously. A total of 240 users completed the experiment. Results were first examined as descriptive statistics and later thrown into a mixed-effects linear regression model to test for significance. The text-only condition came out triumphant, providing not only a boost in accuracy to both Age Groups, but specifically also a disproportionate boost in speed to the older user population.

Introduction

In 2004, 70-98% of websites were found to be inaccessible [?], and one of the user demographics most negatively affected by inaccessible websites is older users [?]. One subject that researchers in the field of Human-Computer Interaction (HCI) study in order to create a more accessible Internet for older users is iconography. Even though semiotics has been a constant throughout human history (via languages), a unique lexicon of icons arose as the Internet developed, defined heavily by Western culture and the newly emerging, globalization-influenced Internet culture [?], making a brand new barrier to entry for older user populations.

Many HCI researchers have looked at whether different variants of Internet icons create positive performance effects for older users, but results have generally trended towards inconclusive. For instance, results varied on whether closer semantic distance, or “the closeness of meaning between icon and function” [? ? ? ?], or adding text labels [? ? ?] positively influences older user’s speeds and accuracies in ways significantly different than younger users.

Most of these studies were also conducted in cognitive psychology-inspired lab studies [?]– they remove the context of the web applications the icons would be embedded into in the real world. There are a couple of north star papers for this thesis that study icon accessibility within a more realistic context. Specifically, Yu & Chattopadhyay provided some groundbreaking results [?]. Yu & Chattopadhyay presented 6 feature-rich mobile app user interfaces (UIs; e.g. Starbucks, AMC, Subway, etc.) and studied whether highlighting relevant features for the participants’s given task would boost performance speeds and accuracies among older adults [?]. Shockingly, Yu & Chattopadhyay found that with emphasized task-relevant features, older adults performed at the same speeds and accuracies during task completion as younger adults [?]. But embedded, task-based environments have been far less explored and do not have the wealth of findings that controlled lab experiment iconography studies have.

Causal findings and realism are trade-offs. Yet, given the large strides in technological advancement being pushed, it’s now more important than ever to tie together real-world applications into accessibility research. Alongside this, studying icons in the context of a realistic, task-based experimental design also could provide greater insight into these generally inconclusive findings.

The goal of my experiment is to measure whether different icon types (specifically text, icons, or icons with text labels) have a positive impact on older users’ performance (as compared to younger users) in a web-based task-completion setting.

This experiment embedded these icons into 6 everyday web-based tasks, such as web browsing and food ordering. The research was conducted online and asynchronously. More specifically, participants were first recruited on Prolific, then redirected to a UX testing platform called Maze where the experiment was conducted, which plugged into Figma, a UI/UX design tool, to access these 6 webpages' designs. Conducting experiments online is a generally new procedure for many social sciences, and although HCI pioneers this push towards asynchronous research, accessibility research within HCI remains a stalwartly in support of in-person studies. With its novelty, a lot of scrutiny can be raised, which is addressed later in this paper in both Methodologies and Results, in sections 3 and 4.

A total of 240 users completed the experiment. Results were first examined as descriptive statistics and later thrown into a mixed-effects linear regression model to test for significance. The text-only condition came out significant, providing not only a boost in accuracy to both age groups, but specifically also a disproportionate boost in speed to the older user population, which is the main finding of this paper. Alongside this, the icon with text labels condition overall performs worse than the text-only and icon-only conditions.

Although creating positive performance effects for both younger and older users is beneficial and should not be written off as completely null and void results, the exciting findings are the ones where older users are boosted in ways disproportionately greater than younger users. This is due to the idea of equity versus equality, where pushing for an equity-based design framework allows older users to access and experience technology in the same ways younger users do.

Chapter 1

Background Literature

In the following chapter, I will discuss the literature motivating my thesis. I will first begin in section 1.1 by discussing the multidisciplinary origins of human-computer interaction (HCI) and how these origins inform HCI’s current-day initiatives. Then section 1.2 will focus in on the HCI subfield of older user accessibility and the standards set for accessibility on the Internet. Section 1.3 dives deeper into the academic research that drives these standards, especially regarding computer iconography, and finally, section 1.4 discusses task-based research involving older users in HCI and summarizes the gap in the literature that my thesis aims to address.

1.1 Human-Computer Interaction

Human-computer interaction (HCI) was born in the 1960s out of the minds of a handful of computer scientists during a time where computers were “scarce, expensive, bulky and formal-scheduled machines used for automatic calculations” [?]. Their goal was to make computers more compatible with humans to “*augment the human intellect*”, according to Douglas Engelbart, one of the founding fathers of HCI [?] (emphasis in original). With this clear focus in mind, graphical user interfaces, mice, applications, and much more were developed [?].

In the 1980s, the tides began to turn as a wave of cognitive psychologists entered the field of HCI. With “[t]he body of knowledge, research findings and methods that made up cognitive psychology...” [?], HCI gained a robust toolkit and a broader, more holistic worldview.

With the 1980’s cognitive psychologists at the helm, the once narrowly-defined goal of HCI quickly expanded, leaving HCI “bursting at the seams” [?]. Fast-forward to current day, HCI research topics includes everything from how individuals interact with their surrounding technology ecosystem (human-environment interactions), ethics in data privacy, alongside technology’s potential in various sectors such as health, social governance, and education [?].

One of the biggest fields that HCI encompasses is accessibility, which “aims to ensure that IT applications and services can be used on an equal basis by users with disabilities and older users” [?]. The need for expanded accessibility research cannot

be overstated, as the U.S. Centers for Disease Control and Prevention estimated that one in four people in the United States live with a disability [?]. Accessibility is a subfield that, even within a very diverse parent field, stands out for its multidisciplinary basis. Beyond computer science and psychology, many other fields also inform HCI's accessibility research, including user experience (UX) design [?], ergonomics [?], biology [? ?], and even economics [?].

1.2 Older User Accessibility

As Stephanidis explicitly mentions, one of the key subjects of HCI accessibility research is older users [?]. After all, the National Institute of Aging states that people over 60 are one of the fastest growing user groups on the Internet [?], and older users quite literally experience the Internet differently than younger users [?]. This is due to age-related decline in vision [? ? ? ? ?], which includes things like declines in color discrimination and narrowing of visual fields, memory [? ? ?], hearing [? ? ?], motility [? ? ? ?], and more. Mahmud et al. also reveal that even the exploration strategies older users employ on technological interfaces are different than other age groups [?]. It should be noted that despite all of the biological developments of aging, older people might not see themselves as “disabled”, which uniquely situates them in the wider accessibility discussion.

Yet despite this overwhelming need for accessibility, according to a 2004 paper, 70-98% of websites are not accessible [?]. In order to ease the burden of encoding accessibility into websites, HCI researchers aim to update web accessibility standards to best match the needs of older users today.

The World Wide Web Consortium's Web Content Accessibility Guidelines (WCAG) are hailed as “the epitome of Web accessibility guidelines” [?]. The WCAG is targeted towards software developers, allowing them to weave accessibility even at the granularity of individual lines of code. Recent amendments to the Americans with Disabilities Act also make the WCAG the definitive standard for any institution receiving federal funding, which extends to governmental organizations, higher education establishments, and more [?]. Other widely-used accessibility standards include Accessible Rich Internet Applications (ARIA), which is derived from UX standards [?], and Section 508 of the U.S.'s Rehabilitation Act (29 U.S.C. 795d), which puts forth legal requirements for accessibility within American information and communication technology [? ? ? ?]. Other countries have also published their own guidelines on web accessibility [?].

1.3 Older Users and Icons

But what informs these accessibility standards? Behind the scenes, HCI accessibility researchers have kept pace with the ever-changing realm of technology, maintaining up-to-date standards even when innovation turns web accessibility on its head. One of these aspects of technology that has undergone substantial development in the mere decades since the inception of the Internet is icons.

Iconography is a constant in human history. Semiotics, which studies icons, understands them as the foundation of human communication systems, with meanings that are socially and culturally constructed [?]. As the World Wide Web (WWW) emerged, its graphical interface naturally brought rise to the WWW's very own lexicon of iconography, defined primarily by Western culture and the newly emerging, globalization-influenced Internet culture [? ? ?]. And as icons have emerged as a new mechanism of expressing meaning on the Internet, whose culture ever-fluctuates, HCI researchers have begun delving into the various facets of this brand new virtual iconography.

1.3.1 Semantic Distance

One of these areas of interest is semantic distance, which is defined as “the closeness of meaning between icon and function” [?]. Results are varied on whether closer semantic distance has a significant positive influence on older users' speeds and accuracy.

Liu et al. studied the impact of semantic distance (concrete versus abstract) and icon design (skeuomorphic versus flat, which is discussed in 1.3.2) on older versus younger participants' reaction times [?]. Specifically, Liu et al. ended up studying four different icon types—skeuomorphic concrete, flat concrete, skeuomorphic abstract, and flat abstract. They aimed to measure the resulting cognitive load of participants, which was quantified via performance measures such as reaction time (RT), physiological measures (pupil diameter and the number of fixations), and subjective assessment measures. Participants were given semantic words, then were asked to judge whether the icon shown after the word matched. After analysis via a 3-factor analysis of variance (ANOVA), they found that when icons had further semantic distance “older participants had significantly higher reaction times... than younger participants” [?].

Dosso and Chevalier also found similar results in their study of semantic distance (close versus far) and text labels (icon versus label versus combination, which will be discussed in 1.3.3) on different age groups (young versus old) [?]. This study measured RTs (both including rereading and without rereading), correct response scores, and the number and length of rereads when presenting users with 30 interfaces of icons. Within the interface, there were 20 icons total, 19 being distractors and 1 being the target, the only study out of the ones in 1.3 that shows 20 icons in an icon selection-based task (most opt for 3x3 arrays, as seen in [? ? ?]). Through ANOVAs performed per dependent variable, Dosso and Chevalier concluded that “older users needed more time to select icons, specifically when semantic distance was far” [?].

However, the sureness of these findings have been contested, such as by Ganor and Te'eni [?] and Schröder and Ziefle [?]. Ganor and Te'eni observed whether differences in an icon's level of detail (simple versus detailed) and semantic distance (close versus distant) created any significant effects on different age groups' (young versus old) performance time and performance accuracy [?]. They achieved this by setting forth 40 tasks, where each task had participants choose the target icon amongst 7 other distractors, which, like Dosso and Chevalier, is an uncommon array

format [?]. Although they did find a greater negative impact of semantic distance on user performance on older participants, this impact was not significant according to their ANOVA analyses.

Schröder and Ziefle also looked at similar independent variables, manipulating icon concreteness and semantic distance (concrete versus abstract) and complexity (simple versus complex) on different age group's (young versus old) RT and fraction of confirmatory responses per item (CR) [?]. This resulted in 4 types of icons, namely abstract simple, abstract complex, concrete simple, and concrete complex. Schröder and Ziefle specifically showed mobile phone icons and gave 4 targets and 4 “dummies” per task for participants to select from— this is the only study in 1.3 that uses more than one “dummy”/distractor icon per task (see [? ?] and more). Through a 3-way repeated measures ANOVA, they found that although there was no significant effect of semantic distance on RT, there was a significant effect on CR, and still argued that concreteness is a more crucial factor than complexity when designing icons for elderly users in an inconclusive manner [?].

Generally, the consensus is that semantically close icons elicit faster reaction speeds, but this has not been confirmed to benefit elderly users more so than the average user in a significant manner.

1.3.2 Skeuomorphic Design

As mentioned in Liu et al., the design of icons have also been investigated extensively with somewhat mixed results [?]. Specifically, skeuomorphic icons are of great interest to HCI researchers. Skeuomorphic icons, or skeuomorphisms, are icons that “are usually designed with 3D effect graphics to imitate real objects through gradients, textures, highlights and shadows, and other visual features to bring users a realistic feeling to help them better understand the interface” [?]. These are usually studied in conjunction with flat icons, both of which you can see in Figure 1.1 below. Liu et al., which as mentioned above had independent variables of semantic distance and skeuomorphic icon design [?], and Wu et al., which studied skeuomorphisms in connection to text labels, among other conditions [?], both found that skeuomorphisms had a positive impact on older users' reaction speeds.

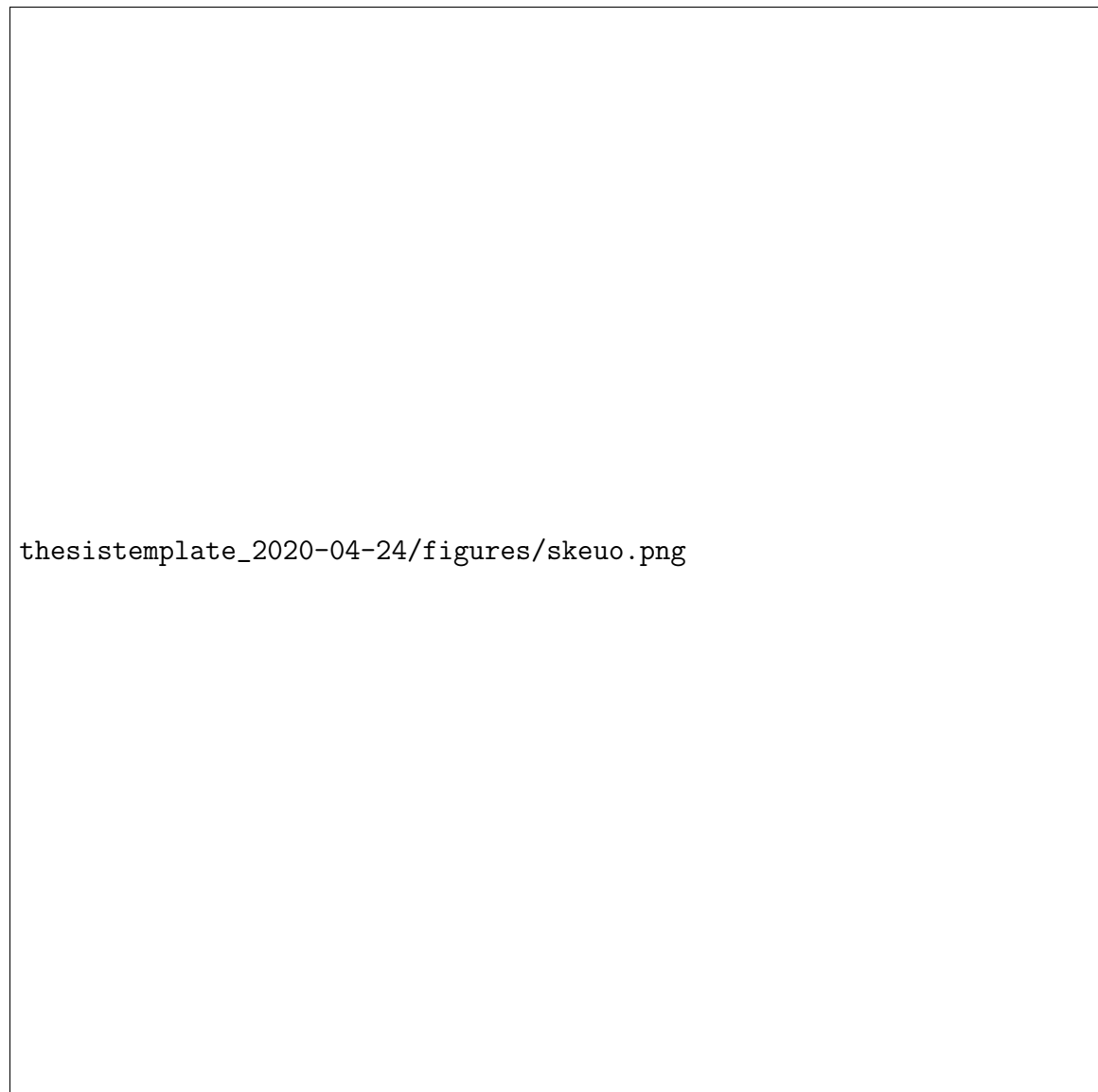


Figure 1.1: From Liu et al. [?], examples of (a) Skeuomorphic Concrete, (b) Flat Concrete, (c) Flat Abstract, and (d) Skeuomorphic Abstract icons (reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com>)

Specifically, Wu et al. looked at icon type (skeuomorphic icon and text versus flat icon and text versus skeuomorphic icon versus flat icon) and cognitive level on search accuracy, RT, and time to first fixation (the time it takes before glancing at the target icon) [?]. Participants were all 60-66 years old and based in Hangzhou. They were asked to look at 3x3 arrays of icons, with 8 distractors and 1 target, all comprised of icons seen in medical services technology in the regions of Beijing, Shanghai, and Hangzhou. Skeuomorphisms elicited the highest search accuracies, fastest RTs, and fastest times to first fixation (with the order of icon types from best to worst being

skeuomorphic icons and text, flat icons and text, skeuomorphic icon, flat icon) [?]. These effects seemed to be the results of reduced semantic distance via realistic design.

But other studies, such as Xi and Wu, found that flat icons, as opposed to skeuomorphic icons, had high search performance within older participant groups [?]. This was found in a study centered around icon style (line versus metro versus skeuomorphism versus flat) and its effect on total visit duration (per icon) and eye movement contrail (the moving line of sight during a task). Notably, this study's participant group were all graduate students, unlike all other studies within 1.3 and 1.4's readings. However, like Wu et al., icons were arranged in a 3x3 grid, with the same ratio of distractors to target icons [?]. After ANOVA analysis, they found that the total visit duration for flat icons was significantly shorter than skeuomorphic icons. Xi and Wu suggested could be due to the higher cognitive load which comes from deciphering the highly-detailed skeuomorphic designs.

Most results point towards skeuomorphic icons over flat icons for elderly-friendly design, however Xi and Wu creates greater nuance in this conversation [?], since positive effects elicited by certain design choices to elderly users are typically also experienced by younger users.

1.3.3 Icons and Text Labels

A subtopic within accessible iconography that might be the most inconclusive as of now is the impact of text labels. Previously mentioned studies such as Dosso and Chevalier [?] and Wu et al. [?], among others, show that older users perform faster when presented icons with text labels as opposed to icons alone. However, Dosso and Chevalier state that this positive impact was experienced equally by both younger and older age groups [?], which is seconded by Song et al.'s findings [?].

Song et al.'s study studied not only text labels, but also icon style (similar, symbolic, example, arbitrary), which mashes together the explorations of semantic distance (1.3.1) and icon design (1.3.2) [?]. Participants of different ages (young versus old) were measured in RT and accuracy in 72 tasks which, like Liu et al., first showed the semantic name, then showed a 3x3 array of 1 target and 8 distractor icons [?], like Wu et al. and Xi and Wu [? ?]. According to a 2-way ANOVA, the RT of older participants increased across all icon types when text labels were provided, and most exciting of all, older and younger accuracies were equivalent after adding text labels to the icons (which without the text labels were not) [?].

Islam approaches this subtopic of elderly accessibility design in a slightly unconventional manner [?]. Islam still conducts a laboratory study, but instead of providing analysis just via quantitative measures, Islam centers their behavioral study around semi-structured interviews. Specifically, after sourcing icons from 2 web domain interfaces, namely a museum webpage and a university webpage, they show the icons to the participants and ask them a series of both open-ended, probing, and closed questions, resulting in interviews between 100 to 120 minutes long. These questions included "What could be the referential meaning [of the icon]?" and "How confident are you that you are correct?" [?]. Islam's participants were largely

graduate students, but not entirely graduate students like [?], with 22/28 being graduate students, 5/28 being full-time company workers, and 1/28 being research assistants. Through qualitative and quantitative analyses, Islam found that “the textual interface signs showed better performance in 4 groups; iconic signs showed better performance in 3 groups; and icon[s] with text sign[s] showed better performance in [the] remaining 4 groups” [?], pointing towards no definite right choice for whether text labels on icons are more accessible for older users.

However, these 3 dimensions of text-label-icon interaction have not been widely studied, so my research hopes to further add to this budding discussion on the impacts of text labels on icon accessibility for older users.

1.4 Older Users and Task-Based Experimental Design

The vast majority of the studies cited in section 1.3 are modeled after traditional cognitive psychology methodology, recording reaction times to stimuli presented in isolation without the context of an actual website surrounding said stimuli [?]. Isolating the icons does reap strong causal conclusions, but in the vast, virtual universe of the Internet, are icons ever shown in isolation? Causal findings and realism are trade-offs, and it is my belief that, given the previously inconclusive laboratory experiments on older users and iconography, studying icons in the context of a realistic, task-based experimental design would provide greater additional insight.

And research on the negative impacts of website clutter on older user performance speeds further this point. Nadkarni in [?] phrases it best: “perceived website complexity (PWC) is central to understanding how sophisticated features of a website... affect a visitor’s experience”. Grahame et al. defines clutter as pixels as a percentage of the available page space that was occupied by text or graphics [?]. This was primarily studied by controlling for the number of links on a webpage, but Grahame et al. also manipulated the font size of the links, the placement of the target link on the UI, etc.. The two age groups (young versus old) were shown web pages modeled off of free use templates and were recorded for number of errors, RT, and fixation number. Through ANOVAs calculated per dependent variable, Grahame et al. determined that increased clutter significantly increased search times and increased error rates for older adults [?].

And although HCI leans towards laboratory studies, there is a longstanding history of task-based procedural research. One of the earliest examples of this can be found in 1993, where the motivation of studying tasks women were “more likely to perform... in the real world” [?], Czaja and Sharit studied how age impacted performance speeds for women, ages 25-75 years old, across 3 different tasks— namely, data entry, file modification, and inventory management. Data entry was simple and repetitive, requiring the participants to change addresses according to instruction cards. File modification was an intermediary level task, where participants were given a database of high-schoolers organized by grade level and asked to check for consistency between

individual student records and instruction cards. Finally, inventory management was the least structured and most demanding task, requiring participants to track the inventory of products from outside suppliers and manipulate this relational database according to instruction cards. Czaja and Sharit showed that increased age, as well as computer experience, were the greatest determiners of faster response times and a smaller number of errors across all 3 tasks [?].

In Mahmud et al., three age groups were studied, each with 10 participants per group: children, adults, and older adults [?]. However, this study takes a different approach than Czaja and Sharit—Mahmud et al. instead asked these 3 age groups, who all were first-time users of Microsoft’s OneNote, to complete 4 main tasks and observed their exploratory styles [? ?]. These tasks were complex, multistep tasks, including making a new notebook, adding relevant notes to a notebook, and sharing notebooks with friends, etc.. After coding video footage of the participants using OneNote and analyzing pre- and post-exploration interviews, Mahmud et al. found a great treasure trove of findings on older users in embedded, task-based exploration settings. For instance, as many of the other readings mentioned may suggest [? ? ?], older participants are overall slower at task completion as compared to other age groups. Their exploration style tended to not fully explore all options, rely on previous experience with other UIs to their detriment, and experience more breakdowns when faced with errors. In fact, older adults typically spent a longer time repeating sequences of failed selections, whereas adults and children more often redirected their efforts based on error messaging. And generally, Mahmud et al. found that older participants were generally dissatisfied with self-exploration, whereas both children and adults maintained high satisfaction levels and confidence before and after the study.

The study that inspired my thesis the most is Yu and Chattopadhyay’s 2024 study of mobile tasks within the context of complex, feature-rich UIs [?]. This experiment involved multiple phases, first beginning with a young (30 participants) versus old (30 participants) comparison of task completion given 4 relevant feature highlighting techniques, where weighted zoom and highlight with context prevailed. The final phase was implementing an AI-assistant to reduce the search space powered by OpenAI, NavNudge, which exhibited 94% accuracy for their *Query* dataset and 93% accuracy for their *UILabel* dataset. Yu presented the UI of a map application none of the participants (who were all older (10 participants), none overlapping with the first phase of the study) were familiar with and studied whether providing an AI assistant (NavNudge) which could decipher speech prompts and highlight (via weighted zoom or highlight with context, as determined by the 1st part of this study) the top 3 most relevant features based on the participant’s given task would boost performance speeds and accuracies among older adults. For example, an example of a task would be “Find out how long it will take to travel from Chicago O’Hare Airport to Chicago Union Station using public transit.” [?]. Specifically, older adults were first presented with the UI without visual cues, and if they remained stuck on the given task for more than 13 seconds or willingly gave up, they were able to use NavNudge to recover. Shockingly, Yu and Chattopadhyay found that with emphasized task-relevant features, older adults performed at the same speeds and accuracies during task completion as younger adults [?]. And it wasn’t that the older

adults were not getting tripped up as easily, it was that with the help of NavNudge, these older adults were recovering successfully, as opposed to the behaviors exhibited by older adults when they encountered errors in Mahmud et al. [?].

If this decomplexification effort can be replicated with icons, whose purpose is to use condensed symbols to represent larger meanings, especially when studied in a task-based experimental design, this would be monumental. As seen in the HCI research above, it is exceedingly rare to find design changes that allow older users and younger users to perform as equals. If substantiated, these findings could introduce a brand new paradigm of iconography and older user accessibility.

Chapter 2

Methodology

This chapter discusses the methodology of this study. As an overview, this study was conducted online and asynchronously. I used a participant matching platform called Prolific for recruitment, and then redirected participants to Maze, a UX usability testing platform, to measure their speeds and accuracies in completing 6 tasks embedded within web interfaces.

In this chapter, Section 2.1 first justifies the use of Prolific for participant recruitment, especially in light of recent developments in artificial intelligence (AI) and its potential harm towards sound online data collection. Then, Section 2.2 runs through the procedure on Prolific and Maze and discusses the design choices behind the Figma UI wireframes used in the procedure. Finally, Section 2.3 discusses the data analysis strategy for the quantitative (speed and accuracy.)

2.1 Prolific and Online Survey Platforms

Prolific, as the introduction to this chapter states, is an online recruitment platform. It allows researchers to post online, asynchronous studies to the platform for its exceptionally large user base to participate in. Surveys can gather large samples in a matter of hours or even minutes. However, the reliability of the platform has seen significant doubt as AI makes rapid gains in its ability to make autonomous decisions and actions.

2.1.1 In Favor of Online Survey Platforms

I begin by introducing a quote from a comparative analysis of Prolific, which at the time was an up-and-coming alternative to the long-used Mechanical Turk (MTurk) from Amazon: “The success of online experiments is not surprising, as they offer at-scale recruitment of participants in a short time, are generally cheap, and offer access to a broader population – potentially even representative of the internet population – than classical lab experiments.” [?]

And these benefits are especially exciting in the elderly accessibility area of HCI, where recruitment of older participants is especially difficult. For instance, Yu and

Chattopadhyay performed a power analysis using one study to estimate the expected effect size, which resulted in a total of 60 participants [?], and Mahmud et al. had only 30 total participants [?] – and as a reminder, these studies are further divided by age group, with Yu and Chattopadhyay having older and younger subgroups and Mahmud et al. having a children/adults/older adults 3-way split. And although the results in these papers are exciting, with such limited sample sizes, it’s hard to ensure the ability to generalize findings and the adequately representative diversity of these samples. This is mitigated with Prolific, which has a total “active, verified, and engaged” participant base of over 200,000 users [?].

“[G]enerally cheap” is also a key benefit, but it’s of note that Prolific pays an ethical wage, unlike other platforms such as MTurk, which is known for “unreasonably low subject payment” [?]. Specifically, Prolific ensures a minimum fixed payment per unit of time, calculated based on a minimum hourly reward of \$8 U.S. Dollars (USD), with a recommended hourly reward of \$12 USD [?].

Online survey platforms such as Prolific are no silver bullet, though. There’s longstanding belief that these participants, in order to make cash out quick, speed through experiments to earn their incentive and move on to the next survey [?]. This fear is only amplified with the onset of AI in an online survey context. Peterson and Asher et al. both studied AI-generated free responses on MTurk and Prolific respectively [? ?]. Peterson found that just 20% of the 1443 responses passed quality checks and counted as usable data [?], whereas Asher et al.’s findings on Prolific was a little more hopeful– when using a keystroke tool to flag submissions with an unusually low number of keystrokes, they found that only 9% were AI-assisted responses [?]. But generally, MTurk is more vulnerable to AI responses [?], partially due to its unethical wages, which encourages long-tail exploitation that oftentimes isn’t valuable enough to humans to engage in [?]. And now, with the rise of commercially-available AI models that can easily execute computer tasks, such as OpenAI’s Operator [?], Claude’s Sonnet 3.0 [?], and Google’s Gemini 2.5 [?], things look even more bleak for online survey models such as Prolific. It is important to note that no real research has been done on the topic of so-called “computer use” AI models participating in online surveys yet.

Luckily, Prolific takes measures against AI and participants who rush through studies. Prolific allows researchers to manually approve study participants for payment, which can help researchers screen for participants with unusually fast completion times or other suspicious activity [?]. The platform, during the writing of this paper, is beta-ing an option to automatically “[r]eject exceptionally fast submissions”, which also deters against these bad actors [?]. But none of these absolutely destroy the threat. In contemplating trade-offs, both via reading the literature presented above and discussing with Reed faculty and staff (see my [Acknowledgments](#)), the ability to seamlessly recruit a notably difficult demographic at such large numbers outweighed the threat of falsified data. If anything, by bulking up the sample size, it mitigates the effect of a singular, or even a handful of AI-responses, if any. But this is not to say that the risk is zero, which is a hard truth kept front-of-mind throughout the following phases of this thesis.

And as briefly touched upon above, another concern lies in whether the results will

be overall skewed faster than average because some online survey users tend to rush through surveys to earn more, and that generally these users will be more fluent in computer technology given that they actively participate in online survey platforms such as Prolific. This concern is valid and real, however, this is a bias that could have just as easily occurred in an in-person study as well. To gather a significant number of older users, one could suggest recruiting them in community spaces elderly folks frequent, such as the library and senior living communities, but people who willingly participant may be more confident and skilled in computer-use than the average older user (although perhaps not to the same extent). And in terms of diversity, online recruitment wins over local recruitment.

All in all, Prolific is not the perfect recruitment tool. But in order to get significant results and to ensure smooth recruitment and data collection, this study has decided to use Prolific to meet its ends.

2.2 Experimental Design

2.2.1 Prolific Recruitment and Screening

Participants were initially recruited via Prolific, the online survey platform discussed in the section above. Prolific screened for participants with the following attributes: 1+ years of web-use experience, conversational English fluency, located in the U.S.A. while participating, using a laptop or desktop to take this survey, and, of course, whether they're a part of either the younger (18-35 years old) or older (60+ years old) age groups.

As a sidebar, within HCI, “old” is fluid and doesn’t refer to a single, well-defined age group— some researchers define “old” by their nation’s retirement ages, whereas others define old age via years in the workforce, and some even rely on local community’s perceptions to inform their definitions [?]. These age groups are inspired by the age groups presented in the previous literature mentioned in chapter 1, which had no clear consensus on what “young” and “old” mean, as seen in Figure 2.1.

Study	Young Age Range (Years)	Old Age Range (Years)	Notes
Liu et al. [?]	22-25	58-69	
Dosso and Chevalier [?]	18-24	60-81	
Ganor and Te’eni [?]	20-35	65+	
Schröder and Ziefle [?]	19-29	50-65	
Wu et al. [?]		60-66	Only studied older users
Xi and Wu [?]			Only studied graduate students
Song et al. [?]	Mean age 22.33	Mean age 67.92	
Grahame et al. [?]	20-31	55-62	
Mahmud et al. [?]	20-35	60-75	Also studied children ages 10-12
Yu and Chattopadhyay [?]	20-29	60+	

Figure 2.1: Table of age ranges in Section 1.3 and 1.4’s readings

And despite the concerns discussed in Section 2.1, it still felt necessary to confirm that they were proficient in computer technology. Since the survey is conducted in English and they would need to complete tasks on English-language website mockups, they had to be competent in English. U.S.A.-based was to fashion culturally-relevant data (and also to trivially comply with the European Union/People’s Republic of China’s data privacy policies), and since the wireframes were websites, limiting devices used to laptops or desktops was intended to make the experience feel more realistic, which was a primary goal of this study as discussed in Section 1.4.

Prolific participants were compensated a base rate of \$1 USD for the roughly 5-minute-long task, from the adjusted \$12/hour Prolific standard wage as mentioned above in Section 2.1, which would be provided after manually screening for abnormal survey results. Once the participants were confirmed not to be bots, they were redirected to Maze for the actual experiment.

2.2.2 Maze and User Tests

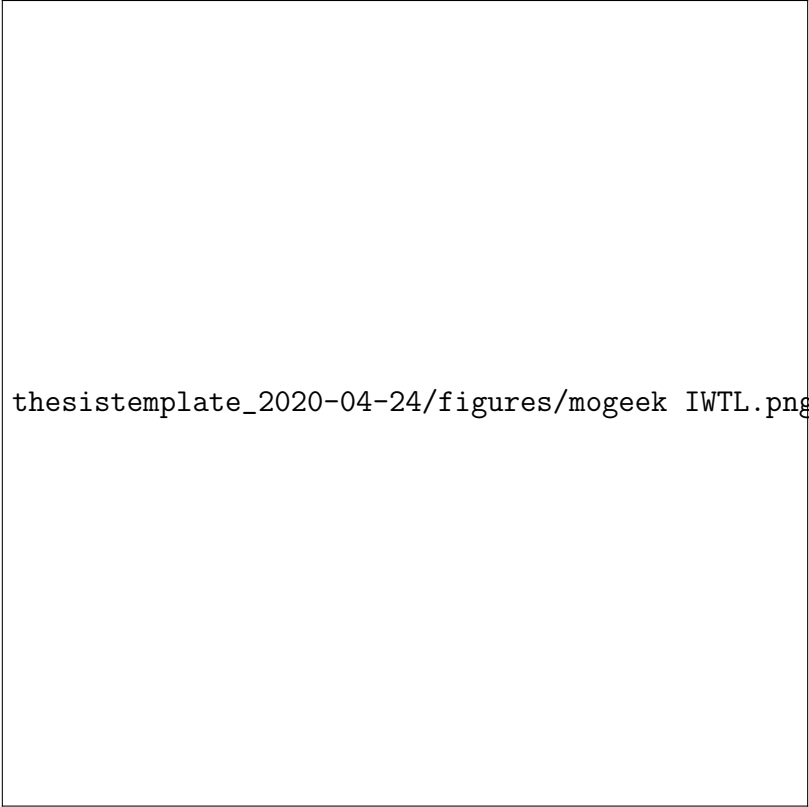
As mentioned above, Maze is a UX testing platform. This means that UI/UX professionals use this platform to test different design choices for their interfaces and record relevant data from participants. There are various platforms of this nature, including Playbook UX, Userbrain, Lookback, etc., but Maze came out on top in terms of pricing, study set-up options, and recorded data types, amongst other features [?]. Maze was also used in a previous Reed Psychology thesis on UX [?], which also informed this choice.

In Maze, participants are once again screened for the attributes listed above in 2.2.1. This adds extra insurance, especially given the concerns in 2.1. They are additionally asked to input their Prolific IDs, which allows for compensation post-experiment. Then, after implying their consent, the meat of the study begins, starting with study instructions. The instructions read as follows:

You will be presented with 6 different websites. For each website, you will be given a task. Please click the correct feature to complete the task. In the case that tasks are complex and require 1+ steps, please click on the correct feature for the first step of the task. You are encouraged to complete these tasks as accurately and quickly as possible.

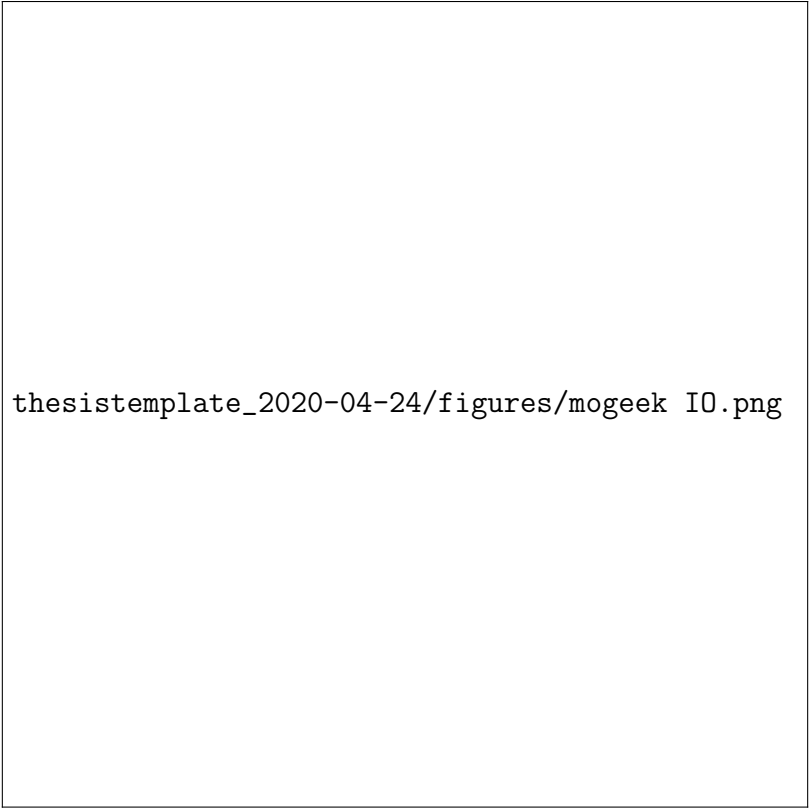
Click continue to proceed.

Then, participants were randomly placed into 1 of 3 groups. Each group was exposed to a different variant of icon type (*Text – Only*, *Icon – Only*, *Icon – with – Text – Labels*); an example of these variants is shown below in Figure 2.1. All groups were exposed to the same six web UIs and tasks in the same order. Specifically, for each of the six tasks, the participants would first see a screen with the task on it, then would have to click to proceed to the web UI to complete the task. As explained in the instructional blurb above, completion of a task would be measured by when the participant clicked on the correct icon to complete the first step of a task. This section of the Maze was achieved by connecting to a Figma prototype, which will be explained more in 2.2.3 below.



thesistemplate_2020-04-24/figures/mogeek IWTl.png

(a)



thesistemplate_2020-04-24/figures/mogeek IO.png

(b)

In the study, the quantitative variables measured were *Speed* (time from when the user enters the web UI the task is conducted on to the clicking of the correct feature) and *MisclickRate* (the number of incorrect clicks divided by the number of total clicks), which is motivated by Yu and Chattopadhyay’s measure of error rate [?].

2.2.3 Figma and UI/UX Design Choices

The six website types were chosen to mimic popular website functions, which was partially inspired by Cloudflare’s Top 100 Domains list [?]. The six chosen website types were: search engine (e.g. Google Search), e-commerce (e.g. Amazon), food ordering (e.g. GrubHub), hotel booking (e.g. Expedia), library (e.g. OverDrive), and video streaming (e.g. YouTube). In order to make realistic web pages that mimicked real-world websites, the wireframes were modeled after real-life websites of these types that are not frequently used by Americans. The “by Americans” caveat is necessary since many of the websites were modeled after relatively popular Australian webpages (namely *Universal Store*, *Fishbowl*, *Vulcan Hotel*, and *Penrith City Libraries*), which were chosen over non-English webpages to avoid translation errors that would render the webpages less intuitive or accessible. The chosen model webpages were *Mojeek* for search engines [?], *Universal Store* for eCommerce [?], *Fishbowl* for food ordering [?], *Vulcan Hotel* for hotel booking, *Penrith City Libraries* for library, and *Gom Player* for video streaming [?]. The different model websites were also chosen to have a variety of perceived complexity, which was qualitatively assessed upon selection. Simply put, complexity was determined by a mix of the number of various buttons on screen, the placement of said buttons, and the difficulty of the task paired with the website UI.

All mockups were created on Figma. Figma is an industry-standard design and prototyping tool used specifically for diagramming technological interfaces (or UIs). It has a wide range of capabilities, including prototypes, which will be discussed later, that allow UI/UX designers to not only mock-up their UIs, but also create interactive elements and functional code [?].

All icons were from Google’s *Material Design Icon Library*, which contains over 2,500 modern, clean glyphs of all meanings [?]. The *Material Design Icon Library* has integrations with Figma as well, allowing the icons to be altered in terms of dimension and color to match the webpages better. This library was also chosen for its use in previous literature such as Song et al. [?].

Most importantly, Figma has prototypes, which allow UI/UX designers to introduce interactive components to their wireframes. In order to create three separate participant groups per variant, there were three prototyped user flows, or orders (a.k.a. “flows”) in which users will experience the Figma mockups. As mentioned in Section 2.2.2, users from all groups first saw the task. Tasks were centered on an otherwise blank Figma frame and would redirect the user to the corresponding web UI where the task would take place. Each participant group saw a different variant of the website. This was achieved by creating three of each task frame that would redirect to the correct variant upon user interaction, which you can see in Figure 2.3

and 2.4. All web UIs were coded such that the users were redirected to the next task frame when the correct icon/feature was clicked. Upon completing the 6th and last task, users were redirected to a blank frame. This blank frame was what Maze recognized as the end of the flow, which would conclude the Figma flow and allow users to continue through Maze. The Figma user flows were directly embedded into Maze.

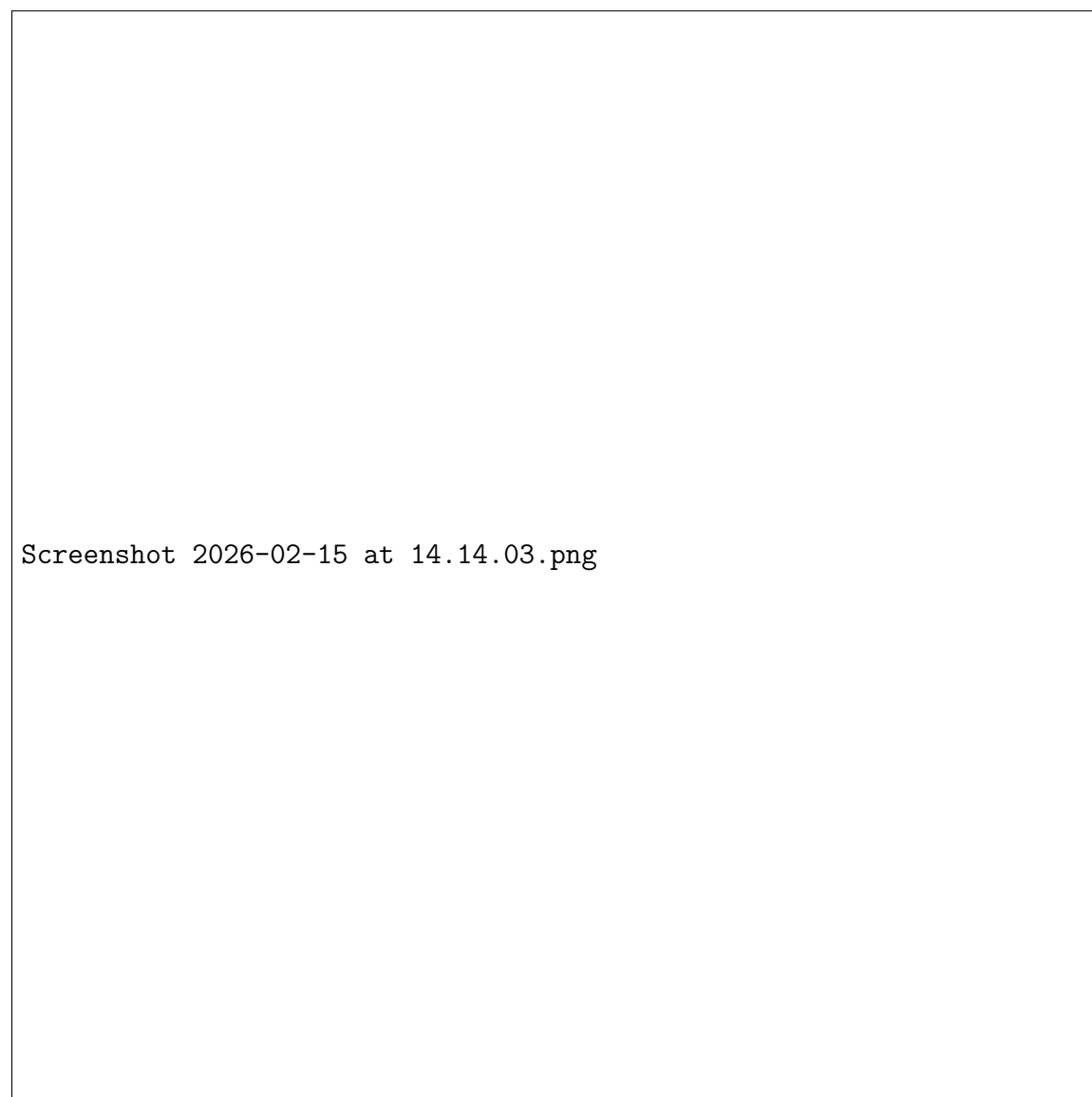


Figure 2.3: Entire Figma file with user flows



Figure 2.4: “Search engine” section of Figma file with user flows

2.3 Data Analysis

2.3.1 Data Recording and Migration

All data was recorded by Maze and hand-replicated within an Excel spreadsheet. The spreadsheet was formatted such that each participant had six entries, one per website. Each entry contained the following information:

1. $Maze_ID$ (a string of integers)
2. $Prolific_ID$ (a string of integers and characters)

3. *Age_{Group}* (either 18-35 or 60+)
4. *Icon_{Type}* (either Text Only, Icon Only, or Icon with Text Labels)
5. *Website* (an integer between 1-6 corresponding to the order of appearance within the experiment's user flow)
6. *Speed* (floating point number, in seconds)
7. *Misclick_{Rate}* (floating point number, represented as a percentage of wrong clicks over total number of clicks)

User responses were only recorded if they had fully completed the study, meaning that they found correct solutions to all six websites. Users who did not complete all six websites or dropped out before being assigned to a condition (which is an important distinction for [later](#) in this section) were not included within the final Excel form.

The spreadsheet ended up containing 1440 total rows, corresponding to 240 unique participant answers. In order to process the spreadsheet information via *pandas*, a Python data analysis library (which will be discussed further in [2.3.2](#) below), the Excel spreadsheet was first converted into a *.csv* file.

2.3.2 Tech Stack

All data analysis was performed in Python, enabled by various Python libraries. The code for my data analysis, as well as the anonymized *.csv* file mentioned in [2.3.1](#), can be found in the Github repository here: <https://github.com/00eemsy/thes>.

The primary Python library employed was *pandas*, which is stylized in all lowercase, is a popular, open-source data analysis library [?]. This was used to access the data within the *.csv* file, clean up the data to remove empty rows and columns, and to perform data manipulation required for both the descriptive statistics and actual linear regression (see below in [2.3.3](#) and [2.3.4](#)). For instance, *pandas* was employed in the descriptive statistics to calculate the average speed of participants per website.

Matplotlib is another popular, open-source Python library used instead for data visualization [?]. All of the data visualizations, from bar charts, histograms, to scatterplots (which can all be found in the Appendix), were constructed by *Matplotlib* with *pandas* data.

Besides these main two libraries, several other libraries were employed in one-off cases, including *scipy* (lowercase intentional) for chi-squared value calculations [?] and *numpy* for array manipulation in tandem with *pandas* [?].

2.3.3 Descriptive Statistics

Performing descriptive statistics analysis before the formal linear regression was necessary to achieve multiple goals. The main goal of these descriptive statistics, besides creating some casual familiarity with the data, was determining exclusion criteria. As mentioned above in [2.1](#), one of the biggest drawbacks to online survey platforms is noisy data. This can be caused by participants speeding through instructions [?],

randomly clicking through UX tests, or even AI bot inaccuracies [? ? ?] (see more in 2.1.1). So cleaning up data is a priority to distill the data points into something potentially of meaning.

The second goal of these descriptive statistics was making sense of the drop-off rate for those assigned an *Icon_Type* variant. As mentioned in 2.3.1, this is related, but distinct, from drop-off rate for those not assigned an Icon Type variant and total drop-off rate. Specifically their relationship is as follows:

$$\text{Total Drop-off} = \text{Drop-off}_{\text{Assigned Icon Type}} + \text{Drop-off}_{\text{Not Assigned Icon Type}}$$

Maze recorded a total of 393 unique attempts, which compared to the 240 successfully completed attempts, was a cause for concern given the $\sim 38.9\%$ total drop-off rate. To this end, descriptive statistics were supposed to give some insight into what could have possibly created this large total drop-off rate, and whether this had something to do either with the areas of interest (mainly, Age Group and Icon Type). Hence, this is why we look specifically at $\text{Drop-off}_{\text{Assigned Icon Type}}$.

In terms of goal one, most of the work was looking at the dependent variables (DV; Speed, Misclick Rate) in relation to the various independent variables (IV; Age Group, Website, Icon Type, etc.). Average Speed was considered both per Website and summed over all 6 websites. To help contextualize these calculations, standard deviation was also derived to understand the variance within these datasets and point towards potential significance. More detailed numbers will be discussed in the Results section below.

This data was then plotted in various manners to help visualize the information. The three main vehicles to visualize this data were bar charts (for visualizing IV/DV correlations), histograms (for plotting distributions of DVs), and scatterplots (to compare between Speed and Misclick Rate).

As for $\text{Drop-off}_{\text{Assigned Icon Type}}$ calculations, the main route for analysis was a chi-square test of independence. A chi-square test of independence decides if two variables are related [?]. This was performed between the number of participants per Age Group and Icon Type who finished the experiment versus those who did not finished but were assigned a variant. To be absolutely clear, those who dropped off before making it to the websites were not included in this measure.

2.3.4 Mixed Effects Linear Regression

Many hypothesis testing methods were considered before settling on a mixed effects linear regression. First, a simpler model of testing was considered, namely a t-test. These are ideal for testing between two conditions, but given that per t-test there is a 5% false positive error rate per test that compounds [?], comparing between more than two conditions (given the three Icon Type variants) was not viable.

One-way ANOVAs were also considered, mainly because of their frequent appearance in the literature [? ? ? ? ? ? ? ?], as a means to analyze data. And to some extent, an ANOVA is simply a type of linear regression. However, ANOVAs perform

mean comparison between categorical variables, whereas a regular linear regression discovers relationships between continuous variable, which was more necessary given that Speed and Misclick Rate are continuous variables.

Finally, the exact formulation of the linear regression has changed over time. Originally, the formula was:

$$Outcome \sim Age + Icon + Website + (Age \times Icon)$$

in quasi-R notation and $DV = \beta_0 + \beta_1(Age_{60+}) + (\beta_2(Icon_{IWTl}) + \beta_3(Icon_{TO})) + (\beta_4(Website_2) + \beta_5(Website_3) + \beta_6(Website_4) + \beta_7(Website_5) + \beta_8(Website_6)) + \epsilon$

This equation measures outcome (either Speed or Misclick Rate) from the main effects of Age Group, Icon Type, Website, and the interaction term between Age Group and Icon Type. However, this misses a key random effect, namely that of Participant, since various aspects of a Participant cannot be controlled for (Age Group, Icon Type, etc.)

The final equation required a mix of both fixed and random effect variables, hence Mixed Effects Linear Regression. In quasi-R-speak, this equation is $Outcome \sim Age + Icon + Website + (Age \times Icon) + (1|Participant)$, which is equivalent to

$$DV_{ij} = \beta_0 + \beta_1(Age_{60+})_i + (\beta_2(Icon_{IWTl}) + \beta_3(Icon_{TO}))_i + (\beta_4(Website_2) + \beta_5(Website_3) + \beta_6(Website_4) + \beta_7(Website_5) + \beta_8(Website_6))_j + u_i + \epsilon_{ij}$$

This is later amended to the following after applying exclusion criteria (which is explained more in 2.3.4):

$$DV_{ij} = \beta_0 + \beta_1(Age_{60+})_i + (\beta_2(Icon_{IWTl}) + \beta_3(Icon_{TO}))_i + (\beta_4(Website_4) + \beta_5(Website_5) + \beta_6(Website_6))_j + u_i + \epsilon_{ij}$$

The u_i represents the Participant random effect. The i indexes an individual participant, whereas j indexes by website, meaning $j \in [1, 6]$. The indices correlate to the two levels of observation, namely the participant level (which include Age Group, Icon Type) and the within-participants level (since each person provides more than one (specifically, 6) outcome.)

With so many terms, overfitting was a risk. Overfitting occurs when terms are added to a model to the point that the model no longer represents the generalizable phenomenon and instead the specific dataset instead [?]. Adding terms always makes for a better fit, but there's a fine line between increasing accuracy in a truly beneficial manner and in an inorganic manner.

In order to avoid this, an additive approach was employed, meaning that each term was added onto the model and then compared while building them up. Model 0 was simply modeling the results, model 1 added in Age Group, model 2 added in Icon Type, model 3 added in Website, and finally, model 4 added in the Age Group/Icon Type interaction term. Furthermore, the performance metric for regression models employed (Akaike Information Criterion) also further negates this risk, which will be discussed later.

In order to compare model fitness, the Akaike Information Criterion (AIC) was employed. AIC for linear regressions is:

$$AIC_p = n \times \ln \left(\frac{SSE_p}{n} \right) + 2p$$

The parameters of AIC are as follows: p is the number of estimated parameters, n is the number of observations, and SSE is the residual sum of squares. Essentially, this statistic estimates the amount of information lost when using a certain model [?]. That being said, the lowest AIC retained the most information, meaning it was the more accurate out of the compared models. The bigger the drops in AIC between models, the more influential that specific effect was in producing the data.

There are several metrics for comparing performance between regression models, including R^2 , but AIC was the best choice because it penalized overfitting and balances goodness of fit and parsimony well [?]. Specifically, the $2p$ penalizes having a larger number of parameters [?], which makes it better for combating overfitting as compared to other statistics such as R^2 .

When calculating AIC, several pairs of models were tested. The difference in AIC between models 1 and 0 was looked at to assess the impact of Age Group on the model. Models 2 and 1 were compared via AIC to estimate Icon Type's influence, models 3 and 2 estimated Website's influence, and finally, models 4 and 3 assess the importance of the Age Group/Icon Type interaction term. Most important for our study was seeing how the interaction between Age Group and Icon Type affected the model's fitness.

After deriving the most fit model, the significance levels (via p-values) were assessed of various variables compared to "baseline" variables (which were chosen by default by *statsmodel*). Through these p-values, true significance was determined.

This Mixed Effects Linear Model approach was employed on a plethora of exclusion criteria, which were determined via descriptive statistics, which are discussed above in 2.3.3. Tests were done with the following exclusions:

1. None
2. Websites 1 and 3
3. Average Misclick Rate > 80%
4. Total Speed > 200s
5. Total Speed > 200s or Average Misclick Rate > 80%

The second exclusion criteria, excluding all results from websites 1 and 3 entirely, was decided upon because websites 1 and 3 consistently performed worse in terms of both DVs (which will be discussed more thoroughly in the Results section). To conjecture, website 3 was a complex, visually busy UI (see below) with a difficult task that relied on clicking a small button ("Remove Mixed Greens from Spicy Salmon order"), making for generally outlying levels of error across both DVs. And website

1, despite having a relatively simpler UI being a search engine (see Figure 2.2), had comparable levels of error possibly due to its appearance as first in the user flow. This was discovered as a possibility during an informal pilot study held before proper data collection performed by advisor Dr. Adam Groce, where he clicked too fast on the first screen and skipped past vital instructions for completion of website 1. If PhD doctors are not immune to it, it is probably a reasonable conjecture to point towards for the general populace of participants.

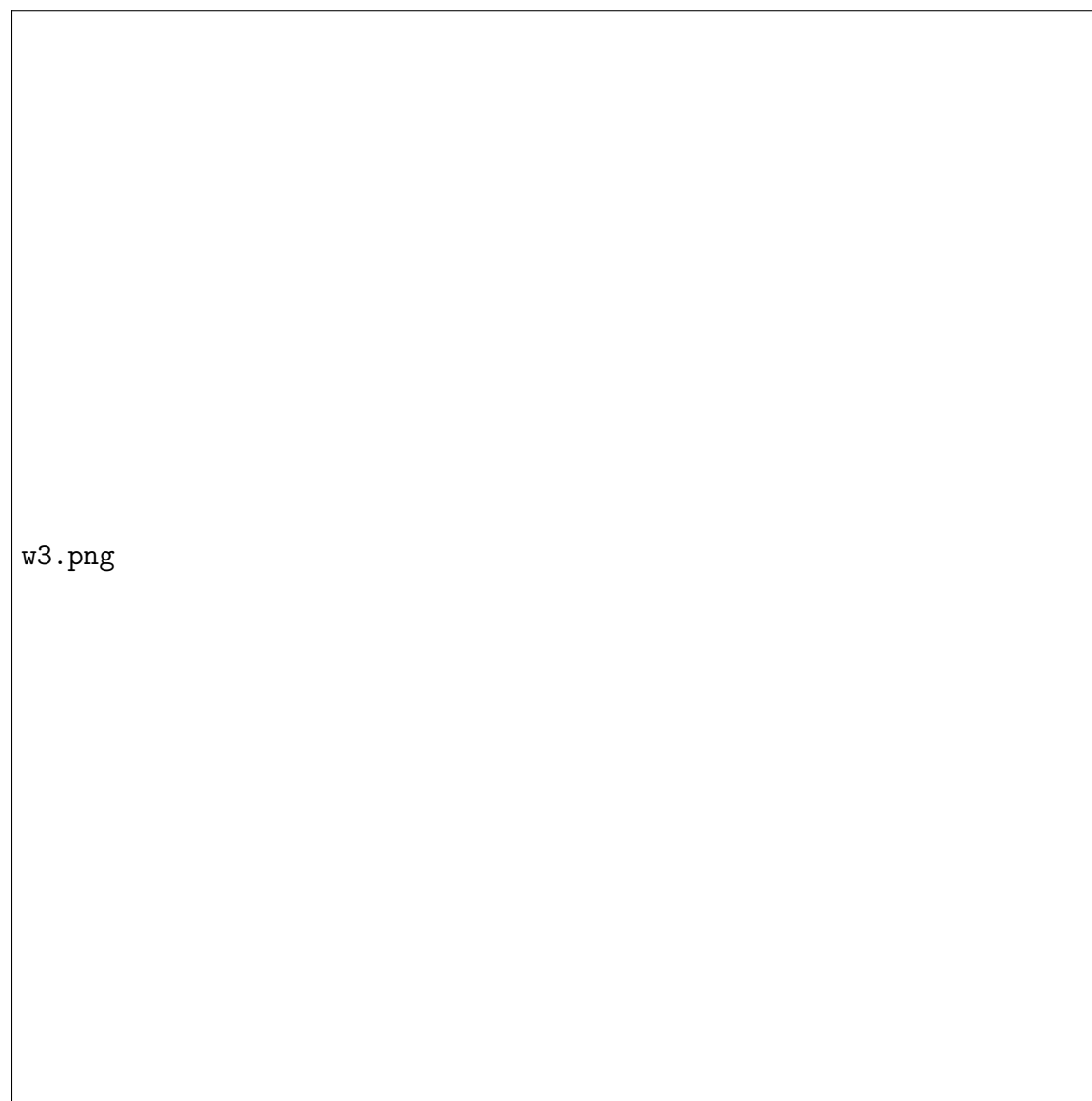


Figure 2.5: Website 3 UI (Text Only variant)

The following scatterplot was the initial justification for utilizing average Misclick Rate and total Speed. Data points usually fell into the lower lefthand corner, which signified faster and more accurate responses. However, a couple scatter far beyond

that larger cluster, which was what created the > 200 s total Speed/ $> 80\%$ average Misclick Rate joint exclusion criteria. The red dotted line in the plot visualizes that cut-off.

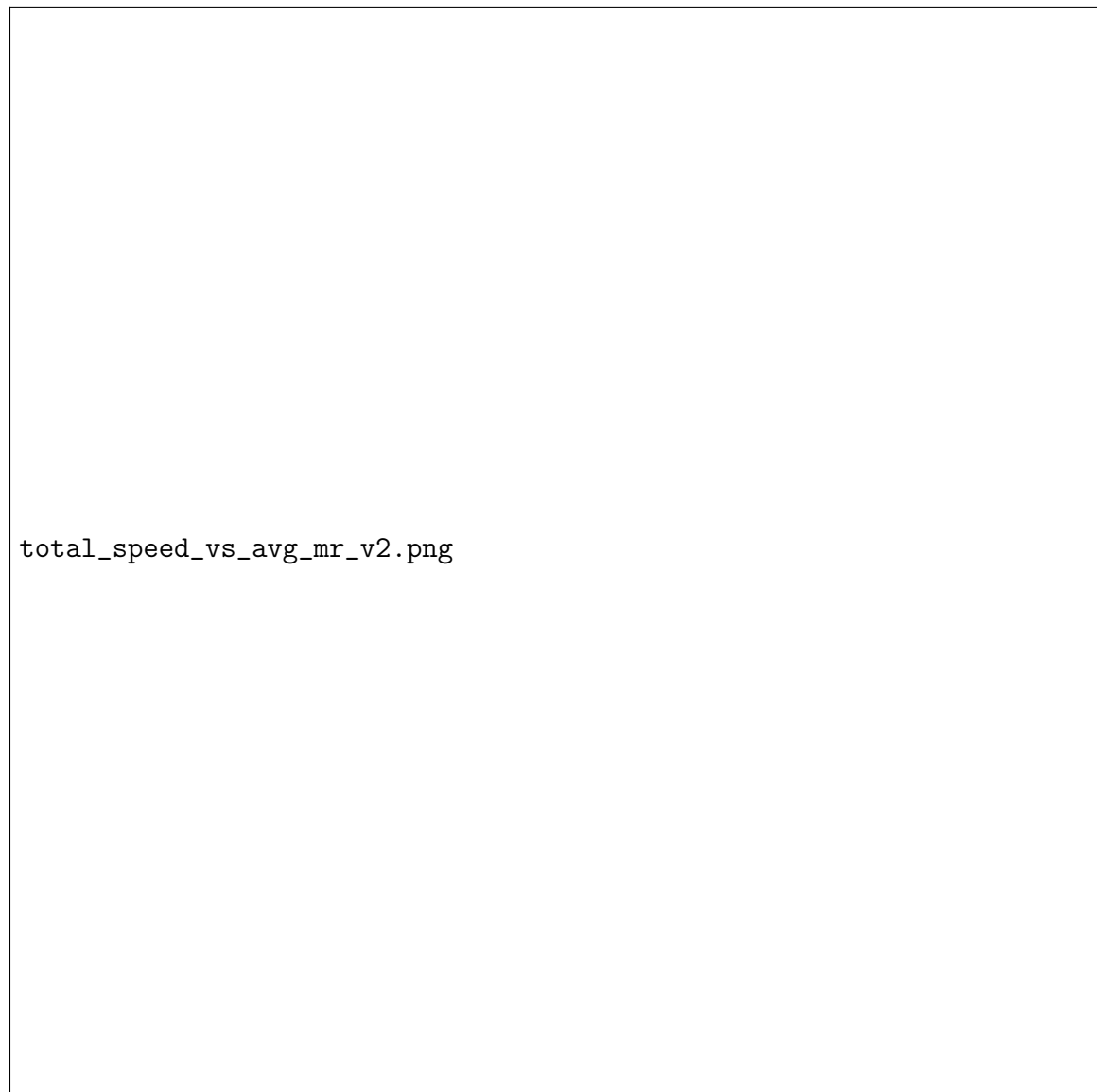
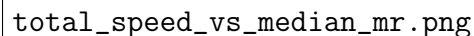


Figure 2.6: Total Speed vs average Misclick Rate by Age Group



total_speed_vs_median_mr.png

Figure 2.7: Total Speed vs median Misclick Rate by Age Group

The use of total Speed and average Misclick Rate was intentional as well. Given the variation in speed per Website, total Speed (the sum of all six speeds across the six different UIs per user) provided a more holistic interpretation of a user's performance. Average Misclick Rate (the average Misclick Rate across all six UIs per user) was also not immune to that variance, and since we were aggregating Speed by participant, there were only a couple of measures we could use for Misclick Rate. For reference, Misclick Rate is calculated as:

$$\frac{\text{Number of incorrect clicks}}{\text{Number of total clicks}}$$

This fraction initially grows fast, but then plateaus early on. This allows for easily

skewed data if even a participant found all websites easy except for one in mean calculations; differences between various magnitudes of errors are also diminished since this function has “diminishing returns”. You can see Misclick Rate values in a table below that illustrate this incredible growth:

# of Wrong Clicks	Misclick Rate (%)
0	0
1	50.000
2	66.667
3	75.000
4	80.000
5	83.334
6	85.714
7	87.500
8	88.889
9	90.000
10	90.909

Figure 2.8: Misclick Rate Growth

Thus, median Misclick Rate (median Misclick Rate across all six UIs per user) was also considered. However, the scatterplot was less clustered and therefore less easily segregated. Hence, average Misclick Rate was chosen. So exclusion criteria 3-5 were all extracted from the total Speed vs average Misclick Rate visualization.

Users with either a total Speed over 200s or an average Misclick Rate of over 80% were completely excluded, even if some of their individual responses had Misclick Rates under 80%.

In the end, the second exclusion criteria of no Websites 1 and 3 was chosen. Reasoning justifying this choice will be presented in 3.1.1, but the reasoning as to why not exclusion criteria 3-5 will be presented now.

In short, the data was noisier than anticipated. Although Figure 2.6 seems like a reasonable cut-off that excludes a lot of noise, the reality is that this excludes a total of 8 users from the data. But with a population of over 200 users and the amount of noise created, removing 8 users’s results was simply not enough to clean the data to make it usable. For reference, the standard deviation for Speed is 16.68296131570999 and the standard deviation for Misclick Rate is 37.17156504702913– standard deviation is variance, which includes both variation due to noise and variation due to signal, but these numbers are particularly high. For instance, many users had good performance on all but one or two Websites, making their average Misclick Rate and total Speed less than the cut-off thresholds but still noisy nonetheless. The main reasoning for not using these exclusions is no proper contextual justification, unlike the exclusion of Websites 1 and 3.

Chapter 3

Results

This chapter will be split into two sections mirroring the split between subsections 2.3.3, Descriptive Statistics, and 2.3.4, Mixed Effects Linear Regression. Each section will give a thorough summary of numeric findings, although it will not cover everything that was deduced— all findings can be found within the Appendix.

3.1 Descriptive Statistics

Descriptive Statistics, as mentioned previously (2.3.3), were performed before the formal linear regression to provide context on a couple of fronts, namely exclusion criteria and drop-off rate. As these were made to define exclusion criteria, the statistics were run on the full set of data without exclusions. The subsequent subsections are organized in lieu of these initial interests, with additional important statistics contextualizing the findings of the linear regression sprinkled in between.

3.1.1 Exclusion Criteria

The final exclusion criteria chosen from those listed in 2.3.4 was the exclusions of Websites 1 and 3 entirely. That meant that the mixed-effects linear regression was performed solely from the data recorded from Websites 2, 4, 5, and 6. This exclusion criteria shaved the initial dataset of 1440 entries down to 960 entries, or exactly two-thirds of the original amount.

The goal of this subsection is to provide justification, both quantitative and qualitative, for this aggressive exclusion criteria. A high-level overview of the argument for this criteria can be found in 2.3.4.

The first statistics helpful in visualizing the validity of this exclusion criteria are [Average Speed per Website](#) and [Average Misclick Rate per Website](#).

Website	Speed (s)
1	17.155250
2	7.857750
3	23.813583
4	7.094208
5	6.433750
6	5.505667

Figure 3.1: Average Speed per Website

Website	Misclick Rate (%)
1	42.977500
2	13.680000
3	64.687917
4	22.932500
5	10.862500
6	9.600833

Figure 3.2: Average Misclick Rate per Website

In both DVs, there are spikes for Websites 1 and 3. Of note is the fact that Website 3 has both worse Speed and Misclick Rate than Website 1 (23.813583s v.s. 17.155250s, 64.687917% v.s. 42.977500%). This backs the original conjecture that Website 3's difficulty was rooted in its UI's complexity and that Website 1's was rooted in the order of appearance. Although some participants, like Dr. Adam Groce (see 2.3.4), perhaps still clicked too fast through Website 1's instructions, this was not a universal phenomenon. This discrepancy points to Website 1's UI not being an issue, and that those who did proceed with caution did not find major difficulty with the website otherwise. However, Website 3's complexity trended more towards universal than otherwise. This can also be seen across Age Groups in tables [Average Speed per Website and Age Group](#) and [Average Misclick Rate per Website and Age Group](#).

Website	Age Group (yr)	Speed (s)
1	18-35	9.126983
	60+	24.665565
2	18-35	4.263707
	60+	11.219919
3	18-35	15.096724
	60+	31.968065
4	18-35	4.294655
	60+	9.713145
5	18-35	4.368362
	60+	8.365887
6	18-35	3.781466
	60+	7.118629

Figure 3.3: Average Speed per Website and Age Group

Website	Age Group (yr)	Misclick Rate (%)
1	18-35	37.704310
	60+	47.910484
2	18-35	7.801724
	60+	19.179032
3	18-35	61.812931
	60+	67.377419
4	18-35	20.285345
	60+	25.408871
5	18-35	4.120690
	60+	17.169355
6	18-35	6.726724
	60+	12.289516

Figure 3.4: Average Misclick Rate per Website and Age Group

No one Age Group finds disproportionate amounts of trouble with Websites 1 and 3 as compared to one another, they are both somewhat affected. This points to experimental design error as opposed to any significant age-related effect.

Variance is also important in contextualizing these findings. Given the hardships posed by Websites 1 and 3, the results reaped from each website happen to be noisier and therefore less useful. This can be seen in [Speed Standard Deviation per Website](#) and [Misclick Rate Standard Deviation per Website](#).

Website	Standard Deviation
1	27.160808
2	8.896048
3	20.878590
4	7.135238
5	8.078409
6	5.290130

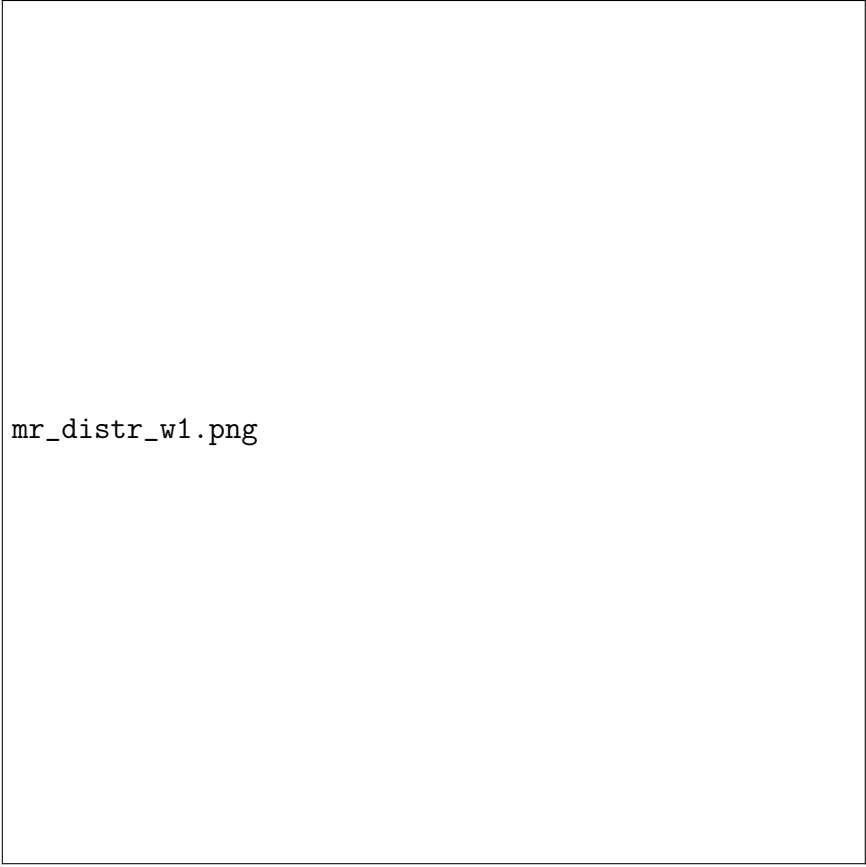
Figure 3.5: Speed Standard Deviation per Website

Website	Standard Deviation
1	38.857985
2	28.404598
3	34.398600
4	33.121365
5	25.779740
6	24.817584

Figure 3.6: Misclick Rate Standard Deviation per Website

In the [Speed Standard Deviation per Website](#) table, it is clearer to see the greater variance in Websites 1 (27.160808, 38.857985) and 3 (20.878590, 34.398600) versus the rest than in Misclick Rate. Misclick Rate, as discussed above in [2.3.4](#), grows fast, meaning the standard deviation in Misclick Rates is inflated by the specific way Misclick Rate is recorded. This can be seen in Website 4's standard deviation for Misclick Rate (33.121365), which is just barely less than that of Website 3 (34.398600), despite Website 4's average Misclick Rate being much lower than Website 3's (22.932500%, 64.687917%).

Some visualizations may help contextualize this inflation. First, consider [Misclick Rate Distribution for Website 1](#) and [Misclick Rate Distribution for Website 3](#). After just 2 erroneous clicks, Misclick Rate shoots over 50% (see Figure [2.8](#)), creating two poles. Notice that in [Misclick Rate Distribution for Website 1](#) the ratio between the number of participants at either end are roughly the same, and that in [Misclick Rate Distribution for Website 3](#) there are more high Misclick Rates than low.



mr_distr_w1.png

Figure 3.7: Misclick Rate Distribution for Website 1



Figure 3.8: Misclick Rate Distribution for Website 3

The distribution for Website 6 (shown in [Misclick Rate Distribution for Website 6](#)) also creates two tails, which is expected given the nature of Misclick Rates (anything between 0-50% is mathematically impossible), but the amount of high Misclick Rates pales in comparison to Websites 1 and 3.



Figure 3.9: Misclick Rate Distribution for Website 6

In sum, this points to “noisier” data from Websites 1 and 3, another case for excluding them from data analysis.

3.1.2 Text Only and 60+

In this next section, I provide some descriptive evidence towards the findings within my linear regression (see 3.2). Namely, that there is a significant relationship between the Text Only (TO) Icon Type and the 60+ Age Group.

As mentioned above in the Background Literature (see 1.3), there is much contention as to whether Icon Only (IO) versus Text Only (TO) variants produce greater performance effects. The following two tables, **Average Speed per Icon Type** and **Average Misclick Rate per Icon Type**, exhibit a neck-and-neck similarity between IO and TO as well.

Icon Type	Speed (s)
Icon Only	10.869219
Icon with Text Labels	12.367133
Text Only	10.793081

Figure 3.10: Average Speed per Icon Type

Icon Type	Misclick Rate (%)
Icon Only	27.064135
Icon with Text Labels	31.788667
Text Only	24.039922

Figure 3.11: Average Misclick Rate per Icon Type

Icon with Text Labels (IWTL) performs worse across both DVs (12.367133s, 31.788667%) than TO and IO, but interestingly enough, TO pulls ahead (10.793081s, 24.039922%) as compared to IO (10.869219s, 27.064135%) in both as well. The difference is small but still apparent, especially when broken down further by Age Group (see [Average Speed per Icon Type and Age Group](#) and [Average Misclick Rate per Icon Type and Age Group](#)).

Icon Type	Age Group (yr)	Speed (s)
Icon Only	18-35	5.954573
	60+	15.661000
Icon with Text Labels	18-35	8.061842
	60+	16.788784
Text Only	18-35	6.481325
	60+	14.370922

Figure 3.12: Average Speed per Icon Type and Age Group

Icon Type	Age Group (yr)	Speed (s)
Icon Only	18-35	23.446581
	60+	30.591250
Icon with Text Labels	18-35	26.070175
	60+	37.661712
Text Only	18-35	19.785897
	60+	27.569858

Figure 3.13: Average Misclick Rate per Icon Type and Age Group

Excitingly, the “best” Icon Type for average Speed between Age Groups differs. The 18-35 Age Group has a faster performance on IO (5.954573s) versus TO (6.481325s), but the 60+ Age Group has the opposite, with an average speed of 14.370922s on the TO condition and 15.661000s on the IO one. Misclick Rate does not exhibit this difference, with both Age Groups erroring less on TO versus IO, but this discrepancy points towards something larger, which is validated in the Mixed Effects Linear Regression (see [3.2](#)). This is expanded upon in the subsequent [subsection on drop-off rate](#) as well.

3.1.3 Drop-Off Rate

As mentioned above in 2.3.3, comprehending the large total drop-off rate ($\sim 38.9\%$) was the second goal of descriptive statistics. Specifically, the question was whether any of our areas of interest (namely, Age Group and Icon Type) had any impact on drop-off rate, specifically the drop-off rate of those assigned Icon Type variants, which is addressed by the following table.

Icon Type	Age Group (yr)	Finishers	Non-Finishers
Icon Only	18-35	39	6
	60+	40	17
Icon with Text Labels	18-35	38	6
	60+	37	23
Text Only	18-35	39	4
	60+	47	20

Figure 3.14: Finishers versus Non-Finishers by Icon Type and Age Group

The number of finishing TO 60+ users (47) is high. This could be in relation to the number of drop-offs with Icon Type assignments versus drop-offs without Icon Type assignments, but a lot of this is obscured away given paywalls set by Maze, the data collection platform.

As mentioned in section 2.3.3, a chi-squared test of independence was run to see if the ratios of finishers versus non-finishers were similar across Icon Types and Age Groups. The returned p-value is ≈ 0.0015 , which makes it significantly different. This makes sense given that decreased ability, both physically and intellectually, is caused in part by increased age (see 1.2). Most of it is most likely rooted in Age Group disparity.

To be completely sure, another round of Chi-Squared Tests of Independence were run, one with only the 18-35 ratio of finishers to non-finishers and another with only the 60+ ratio. Both tests resulted in nonsignificant p-values (0.7875462343821208 for 18-35, 0.5148579660895773 for 60+), which points to the fact that the significant difference was due to differences between Age Group. But it could also be the case that the nonsignificant p-value could have been due to there not being enough data to draw firm conclusions.

3.2 Mixed Effects Linear Regression

The exclusion criteria settled upon was excluding results from Websites 1 and 3 entirely, as justified in 3.1.1. The following linear regression results were calculated off the dataset with exclusion criteria applied.

As aforementioned in 2.3.4, linear regression models were computed in an additive manner, meaning that variables were inserted into the equation one by one to see whether they made a marginal impact on the fitness of the Linear Regression. The following table defines what each model version measures. Note that each

model is grouped by Participant, hence the Participant random effect (written as $(1|Maze ID)$, which was what was used to track unique users).

Model	Definition
Model 0	$Speed \sim 1 + (1 Maze ID)$
Model 1	$Speed \sim Age Group + (1 Maze ID)$
Model 2	$Speed \sim Age Group + Icon Type + (1 Maze ID)$
Model 3	$Speed \sim Age Group + Icon Type + Website + (1 Maze ID)$
Model 4	$Speed \sim Age Group + Icon Type + Website + (Age Group \times Icon Type) + (1 Maze ID)$

Figure 3.15: Definitions of Linear Regression Models

This impact was measured with an Akaike Information Criterion (AIC) to determine the model of best fit. Specifically, the model with the lowest AIC fits the best.

3.2.1 Speed

The AIC for Speed is shown below:

Model	Akaike Information Criterion
Model 0	6522.543660965796
Model 1	6457.99778589942
Model 2	6438.843481013769
Model 3	6427.435013923644
Model 4	6422.526364301187

Figure 3.16: AIC for Models on Speed

Model 4, or the full model, $Speed \sim Age Group + Icon Type + Website + (Age Group \times Icon Type)$, performed the best.

To break it down even further,

1. **Age Group** (Model 1 - Model 0): -64.5458750664
 - (a) Age Group had the largest drop, and therefore the largest impact on results. This makes sense given the literature (see 1.2).
2. **Icon Type** (Model 2 - Model 1): -19.1543048857
 - (a) Still retaining more information than previously, but less influential versus Age Group.
3. **Website** (Model 3 - Model 2): -11.4084670901

- (a) Least impactful so far, but still adding to our model positively! This, as compared to the Linear Regression run without exclusions, yields vastly different results because of how stark the difference between the results of Websites 1 and 3 versus the rest.

4. Age Group x Icon Type (Model 4 - Model 3): -4.90864962246

- (a) Least impactful overall, but still helping the model gain information! This points towards one Age Group performing much better or worse Speed-wise given a certain Icon Type variant.

With this in mind, we run Model 4 (see the [table of results](#) and linear model below). Note that like Linear Regressions, not all variants are represented. The coefficients provided are comparing that variant to a randomly selected baseline variant. Age Group's baseline is 18-35, Icon Type's is IO, Website's is Website 2, and Age Group x Icon Type is 18-35 and IO.

	Coefficient	z	P > z
Intercept	5.050	6.790	0.000
Age[60+]	5.951	6.491	0.000
Icon[IWTL]	1.072	1.155	0.248
Icon[TO]	-0.264	-0.287	0.774
Website[4]	-0.764	-1.130	0.190
Website[5]	-1.424	-2.444	0.015
Website[6]	-2.352	-4.037	0.000
Age[60+]:Icon[IWTL]	0.544	0.414	0.679
Age[60+]:Icon[TO]	-3.018	-2.371	0.018
Group Variance	6.417		

Figure 3.17: Results of Speed Model 4

The results shown in [Results of Speed Model 4](#) can be modeled as follows:

$$\begin{aligned}
 \text{Speed} = & 5.050 + (5.951(\text{Age}_{60+})) + (1.072(\text{Icon}_{IWTL}) - 0.264(\text{Icon}_{TO})) + \\
 & (-0.764(\text{Website}_4) - 1.424(\text{Website}_5) - 2.352(\text{Website}_6)) + \\
 & (0.544(\text{Age}_{60+} \times \text{Icon}_{IWTL}) - 3.018(\text{Age}_{60+} \times \text{Icon}_{TO})) + 6.417
 \end{aligned}$$

There are a couple of things to note from this. First, and most importantly, there is a significant interaction between the 60+ Age Group and the TO Icon Type (p-value = 0.018). This is not only significant, but also a positive effect that shaves off just over 3 seconds on Speed (-3.018). Thus, the effect of TO on the 60+ Age Group, as compared to 18-35, is stronger in the positive direction.

Some other variables were significant as well. Age Group was to be expected (p-value ≈ 0.000), as noted previously, adding nearly 6 seconds to Speed (5.951). This

also puts our main finding into context— for a user in the 60+ Age Group, viewing the TO variant nearly halves the negative performance effects of age.

Interestingly enough, Websites 5 (p-value=0.015) and 6 (p-value $\approx$ 0.000) are also significantly different than Website 2. This might be explained with the same logic that resulted in the exclusion of Website 1. Namely, that learning effect kicks in and allows users to complete later Websites more efficiently due to repeated exposure. This is further backed by the fact that the Speed boost increases between Websites 4 (-0.764) to 6 (-2.352), and that only Websites 5 and 6 are significant as compared to Website 4 (p-value=0.190).

Looking at Icon Types, some interesting trends arise. The different signs of the coefficients mirror what was shown in 3.1.2— that is, that IWTL (5.951) looks to perform worse than both IO and TO, and that TO somewhat performs better than IO (-0.264). Of course, neither of these values are significant, but this changes when looking at Misclick Rate instead of Speed.

3.2.2 Misclick Rate

Again, we begin by calculating AIC (see below).

Model	Akaike Information Criterion
Model 0	9121.182184951773
Model 1	9108.807437991436
Model 2	9083.299844112014
Model 3	9048.092978443245
Model 4	9050.568425915011

Figure 3.18: AIC for Models on Misclick Rate

1. **Age Group** (Model 1 - Model 0): -12.3747469603
 - (a) Much less of an effect than in Speed. This could be do to the way Misclick Rate grows and its two-tailed nature.
2. **Icon Type** (Model 2 - Model 1): -25.5075938794
 - (a) Unlike Speed, Icon Type has more of an impact than Age.
3. **Website** (Model 3 - Model 2): -35.2068656688
 - (a) Again, unlike Speed, Website happens to be the most impactful variable. This could be due to the reasons listed above and also the learning effect explained above in 3.2.1.
4. **Age Group x Icon Type** (Model 4 - Model 3): +2.47544747177
 - (a) Unlike Speed, the interaction term does not increase information retained and in fact worsens our model. It should be of note that this is very slightly different than the AIC for Model 3.

This means that Model 3 is our model of best fit, which denies that the Age Group x Icon Type interaction term helps with model accuracy. This points to that interaction not being significant, which is further validated in the following tables ([Results of Misclick Rate Model 3](#), [Results of Misclick Rate Model 4](#)):

	Coefficient	z	P> z
Intercept	9.903	3.856	0.000
Age[60+]	9.318	4.340	0.000
Icon[IWTL]	6.255	2.337	0.019
Icon[TO]	-8.351	-3.225	0.001
Website[4]	9.252	4.085	0.000
Website[5]	-2.817	-1.244	0.214
Website[6]	-2.352	-4.037	0.000
Group Variance	121.745		

Figure 3.19: Results of Misclick Rate Model 3

	Coefficient	z	P> z
Intercept	10.567	3.533	0.000
Age[60+]	8.007	2.150	0.032
Icon[IWTL]	3.678	0.975	0.330
Icon[TO]	-7.813	-2.085	0.037
Website[4]	9.253	4.085	0.000
Website[5]	-2.817	-1.244	0.214
Website[6]	-4.079	-1.801	0.000
Age[60+]:Icon[IWTL]	5.190	0.972	0.331
Age[60+]:Icon[TO]	-0.886	-0.171	0.864
Group Variance	120.000		

Figure 3.20: Results of Misclick Rate Model 4

Model 3, the best linear model out of the group, can also be represented as:

$$\begin{aligned} \text{Misclick Rate} = & 9.903 + (9.318(\text{Age}_{60+})) + (6.255(\text{Icon}_{IWTL}) - 8.351(\text{Icon}_{TO})) + \\ & (9.252(\text{Website}_4) - 2.817(\text{Website}_5) - 4.079(\text{Website}_6)) + 121.745 \end{aligned}$$

As shown in Model 4, neither interaction term was significant (p-values=0.331, 0.864). So unlike Speed, Age Groups do not have a significantly different Misclick Rate when provided different Icon Type variants.

But, in Model 3, all Age and Icon Types coefficients are significant (p-values=0.000, 0.000, 0.019, 0.001). Specifically, the 60+ Age Group has a marked increase in Misclick Rate versus 18-35 (9.318), IWTL increases Misclick Rate as opposed to IO (6.255), and TO decreases Misclick Rate as opposed to IO (-8.351).

This means that separately, these variables have a significant impact on Misclick Rate. But when given certain Icon Types, the changes in Misclick Rate does not differ strongly between Age Groups the same way Speed does. Again, this could be due to the way Misclick Rate grows rapidly. You also see the impacts of this growth rate in the size of the coefficients, as compared to those in the Speed model.

As for Websites, the only significant difference was between Websites 4 and 2 (p-value $\approx$ 0.000). Not only that, but Website 4's Misclick Rate actually worsens as compared to 2's (9.252). This differs from Speed in a couple of senses— first, Websites 5 and 6 were significant in Speed. But second, all Websites exhibited a positive boost to Speed, whereas Website 4 is performing worse in terms of errors. Even harkening back to Figure 3.2, there is a small uptick in Misclick Rate on Website 4 (22.932500%), albeit nowhere as large as Websites 1 (42.977500%) and 3 (64.687917%). To conjecture, this could be the result of Website 4 appearing right after Website 3, arguably the most difficult, complex, and therefore frustrating UI of the batch. If Website 3 caused participants to give up and begin clicking around randomly, this frustration could have carried over to Website 4 without them realizing how much more straightforward the task was. Mahmud et al. talks at length about how their older users were much more easily frustrated than their younger age groups, making them repeat unhelpful actions without adjusting themselves [?].

3.3 Robustness Checks

In the following section, the results of the Linear Regression discussed above in 3.2 are compared to these regressions performed on similar, alternative exclusion criteria.

After settling upon the exclusion criteria of no Websites 1 and 3, the alternative criteria to confirm robustness were:

1. None
2. Website 1
3. Website 3

The other options for exclusion criteria mentioned in 2.3.4 were also tried before running regressions, but deemed not as relevant to the current criteria. These results, however, can still be found in the Appendix.

3.3.1 No Exclusions

Again, following the format of section 3.2, AIC results will first be shown, followed by relevant Linear Regression results.

First, we discuss Speed. Model 3's AIC score was the lowest at 11813.533233944247. The model of best fit is the one including all terms except the Age Group and Icon Type interaction term, meaning these interactions terms did not provide helpful information retention for the model. Interestingly enough, AIC actually increased when the variable representing Icon was folded into the equation, meaning it actually did

worse than just Age by itself. Concerning, but nevertheless we move towards the results themselves for more conclusive findings. The following are snippets from the results from [models 3](#) and [4](#) that are most relevant for comparison.

	Coefficient	z	P> z
Icon[IWTL]	1.612	1.343	0.179
Icon[TO]	-0.428	-0.369	0.712

Figure 3.21: Results of Speed Model 3 with No Exclusions

	Coefficient	z	P> z
Icon[IWTL]	2.107	1.234	0.217
Icon[TO]	0.527	0.310	0.756
Age[60+]:Icon[IWTL]	-0.979	-0.405	0.685
Age[60+]:Icon[TO]	-1.817	-0.776	0.438

Figure 3.22: Results of Speed Model 4 with No Exclusions

In model 3, Icon Types had no significance. IWTL actually was closer to being significant than TO (p-values=0.179, 0.712), but neither were particularly close to significant levels. As a note, all other terms (Age Group, Websites) exhibited p-values of 0.000.

In model 4, neither Icon Types nor the interaction term exhibited any significance. The coefficients for these terms are outliers from what we see throughout the main regression (see [3.2](#)) and the rest of the robustness check results. Both IWTL and TO are positive (coefficients=2.107, 0.527), meaning that overall both performed worse than IO. This seems to contradict calculations for [Average Speed](#), but this could be due to model error. However, both interaction terms had negative coefficients (coefficients=-0.979, -1.817), implying better performance from the 60+ group than the 18-35 group in response to these specific Icon Types. This also seems to contradict findings presented in [3.1](#), but again this could be due to model error.

Again, IWTL got the closest out of the four terms to significance (p-value=0.217). As another note, all other terms not listed (Age Group, Websites) exhibited p-values of 0.000.

As for Mischick Rate, model 3 once again outperforms the rest with an AIC of 13927.39569189007, meaning the interaction terms do not benefit model fitness much. Onto selected results:

	Coefficient	z	P> z
Icon[IWTL]	4.838	1.852	0.064
Icon[TO]	-3.376	-1.337	0.181

Figure 3.23: Results of Mischick Rate Model 3 with No Exclusions

	Coefficient	z	P > z
Age[60+]	7.145	1.963	0.050
Icon[IWTL]	2.624	0.712	0.477
Icon[TO]	-3.661	-1.000	0.317
Age[60+]:Icon[IWTL]	4.447	0.853	0.394
Age[60+]:Icon[TO]	0.639	0.127	0.899

Figure 3.24: Results of Misclick Rate Model 4 with No Exclusions

In model 3, IWTL eeks towards significance (p-value=0.064), but still doesn't hit the mark. Again, all other coefficients had p-values of 0.000.

In model 4, coefficients behave more in line with expectations than when looking at Speed (coefficients=2.624, -3.661, 4.447, 0.639), but still nothing of significance. Interesting to note is that Age is barely no longer significant (p-value=0.050), meaning this model did not notice significantly different error rates between the Age Groups. All website terms had p-values of 0.000.

Between both Speed and Misclick Rate on no exclusions, the jarringly inaccurate model reportings showcase how deeply noisy this data was. This was expected, as detailed in 2.1.1, but the **wrongly signed coefficients** and **nonsignificance of Age** nail this point home.

3.3.2 Website 1

As a reminder, Website 1 was initially excluded alongside Website 3 because it was first in the experiment for all users, and this made it prone to a learning curve.

Once again, the lowest AIC for Speed was model 3 (AIC=9119.597747044225), meaning that the interaction terms did not add significant benefits to model fitness.

Models 3 and 4 reaped the following results:

	Coefficient	z	P > z
Icon[IWTL]	1.894	2.058	0.040
Icon[TO]	-0.430	-0.483	0.629

Figure 3.25: Results of Speed Model 3 with no Website 1

	Coefficient	z	P > z
Icon[IWTL]	2.217	1.712	0.087
Icon[TO]	0.883	0.686	0.493
Age[60+]:Icon[IWTL]	-0.626	-0.342	0.733
Age[60+]:Icon[TO]	-2.482	-1.398	0.162

Figure 3.26: Results of Speed Model 4 with no Website 1

In model 3, the most shocking breakthrough is the significance of IWTL (p-value=0.040). Its coefficient of 1.894 points towards IWTL being worse for speed than IO in general.

However, this doesn't hold in model 4. Model 3 has more accuracy in fit, so that explains part of it. But again, the signs on the coefficients are not to expectation. Icon Type terms both have positive signs (coefficients=2.217, 0.883), meaning both perform worse than IO on Websites 2-6. But the interaction terms are both negative (coefficients=-0.626, -2.482), meaning older users perform better given these conditions versus the younger participant pool. But both the Icon Type and interaction terms do not have significance (p-values=0.087, 0.493, 0.733, 0.162).

As for Misclick Rate, once more model 3 outperforms the rest of the batch (AIC=11460.732933369016). So once more, we look towards both models 3 and 4 for insight.

	Coefficient	z	P > z
Icon[IWTL]	7.990	3.103	0.002
Icon[TO]	-2.790	-1.120	0.263

Figure 3.27: Results of Misclick Rate Model 3 with no Website 1

	Coefficient	z	P > z
Icon[IWTL]	5.998	1.650	0.099
Icon[TO]	-3.010	-0.833	0.405
Age[60+]:Icon[IWTL]	4.000	0.778	0.437
Age[60+]:Icon[TO]	0.507	0.102	0.919

Figure 3.28: Results of Misclick Rate Model 4 with no Website 1

We see that IWTL remains significant in model 3 (p-value=0.002), matching Speed's similar findings. Once again mirroring Speed, model 4 does not share this finding. In fact, none of the Icon Type and interaction terms hold significance (p-value=0.099, 0.405, 0.437, 0.919). IWTL is the closest out of the four terms to significance (p-value=0.099), though.

To recap, IWTL seems to have significance when looking just at Websites 2-6. If the exclusionary criteria reasoning was correct, then Website 1's main issue was being placed first in the user flow. If that's the case, errors could have been distributed in a way that obscured IWTL's speed-inhibiting effects, as it does not rely on the Icon Type encountered if instructions were unclear or zoomed past.

3.3.3 Website 3

To recap, Website 3 was initially excluded due to the complexity of the task. The UI was crowded by various design elements (see Figure 2.5), which is necessary for a food ordering webpage. And the success criteria was clicking a small button amongst

the visually busy UI. It's deep enough within the sequence of Websites that it does suffer from the learning curve that Website 1 does, but its generally high difficulty level of a task created a lot of noise in its findings.

First, we look at Speed. The lowest AIC belongs to model 3 (AIC=9634.502880155569). This follows a similar trend with the rest of the robustness check's AICs.

Now, looking at results from models 3 and 4, we get:

	Coefficient	z	P> z
Icon[IWTL]	1.103	1.032	0.302
Icon[TO]	-1.585	-1.534	0.125

Figure 3.29: Results of Speed Model 3 with no Website 3

	Coefficient	z	P> z
Icon[IWTL]	1.169	0.776	0.438
Icon[TO]	-0.463	-0.309	0.757
Age[60+]:Icon[IWTL]	-0.114	-0.053	0.958
Age[60+]:Icon[TO]	-2.112	-1.023	0.306

Figure 3.30: Results of Speed Model 4 with no Website 3

None of the Icon Type terms are significant according to model 3, but it is worth noting that this time, TO's closer to significance than IWTL, as was seen in excluding just Website 1 (see [above](#)).

Again, none of the Icon Type or interaction terms were significant either. But again, the closest to significance was the interaction term for 60+/TO, which was not the case for excluding just Website 1 (it was the second closest [p-value=0.162] after IWTL [p-value=0.087]). As a sidebar, all other terms in both model 3 and 4 were significant.

AIC was no different for Misclick Rate; model 3 still remained the model with the lowest AIC at 11526.779337953953.

As for results, we see that it mirrors a lot of what Speed found:

	Coefficient	z	P> z
Icon[IWTL]	2.820	1.013	0.311
Icon[TO]	-7.942	-2.951	0.003

Figure 3.31: Results of Misclick Rate Model 3 with no Website 3

	Coefficient	z	P> z
Icon[IWTL]	0.092	0.024	0.981
Icon[TO]	-7.634	-1.959	0.050
Age[60+]:Icon[IWTL]	5.488	0.989	0.323
Age[60+]:Icon[TO]	-0.449	-0.084	0.933

Figure 3.32: Results of Misclick Rates Model 4 with no Website 3

Excitingly, TO holds significance in model 3 (p-value=0.002). IWTL, unlike in the models excluding only Website 1, was not (p-value=0.087). Model 4 also mimics this trend with TO, although TO is barely nonsignificant (p-value=0.005).

Excluding Website 3 brings to light the positive effect of TO, even at times the positive effect it specifically has on the 60+ Age Group. This complements the results of excluding solely Website 1, which only showed the negative effects of IWTL. Perhaps TO was able to shine through because complexity of the task surpassed any one Icon Type's ability to positively influence performance metrics.

To summarize, these robustness checks point towards the main results, but do not completely validate them. Without Website 1, the significantly negative impacts of IWTL (p-value for Speed=0.040, p-value for Misclick Rate=0.002) that are mirrored in the main results shine through. Without Website 3, TO's positive impact was signaled towards (p-value for Misclick Rate=0.003). Neither displayed similar significant interactions between the 60+ Age Group and the TO Icon Type. Throughlines are drawn, but the main results do not hold in terms of robustness. Alternatively, these main results are sensitive to specific Websites, namely 2, 4, 5, and 6. This should be kept at front of mind when reading the [Discussion](#) portion of this thesis.

Chapter 4

Discussion

This experiment was intricate and littered with learning many curves for the Computer Science-trained thesis author. That being said, there is a lot to discuss.

Generally, the format of this chapter will be as follows: first, qualitative conjectures will be attached to quantitative results found above in [chapter 3](#), a section that will be labeled as Interpretations. Second, limitations will be brought to light, confronted, and acknowledged. Then, future directions will be suggested, with a resounding conclusion to end this paper.

4.1 Interpretations

Two main results follow from [Results](#), namely that the Text-Only condition has a disproportionately positive influence on the 60+ Age Group (and that TO overall performs better across Age Groups than IO and IWTL), and that the Icon with Text Labels condition overall performs worse than TO and IO conditions. A reminder that these results hold for Websites 2, 4-6, which will be discussed later on in this chapter.

4.1.1 Text-Only and 60+ Interpretations

To reiterate, when modeling Speed, the best model of fit was the regression that included Age Group x Icon Type interaction terms. Within the calculations for this model, the interaction term between the 60+ Age Group and the TO Icon Type conditions had a coefficient of -3.018 and a p-value of 0.018. In other words, the 60+ Age Group averaged 3.018 second less on tasks that were shown in the TO condition than the 18-35 group.

As for Misclick Rate and TO, in both semi-complete model 3 and fully-expanded model 4, TO by itself had significance (p-values=0.001, 0.037) both with negative coefficients (coefficients=-8.351, -7.813). The interaction term from above was not significant.

These findings, especially the significance of the TO/60+ interaction in relation to Speed, are huge. As explained in the Background Literature, one of the main focuses of the age-based accessibility subfield of HCI is finding ways to even the technological

playing field for older users experiencing age-related physical and mental declines. Of course, finding accessibility fixes that both Age Groups benefit from is also exciting, and in practice is what will be employed in building technology but the grand prize is discovering tweaks that overall benefit all ages but that older users happen to also respond better to than younger users. It's in line with notions of equality versus equity— one makes sure all are given equal resources, and the other ensures all are given adequate resources to even the playing field.

This result finds backing in scholarship, as well. As mentioned in , the closer the semantic distance icons are to what they represent, the faster participants are. And what is more semantically close than language itself? However, there is debate as to whether semantic distance provides a larger performance boost to older users versus younger. Liu et al. found that further semantic distance disproportionately hindered the reaction times of older participants compared to younger participants [?]; Dosso and Chevalier also found similar results [?]. But these age-based discrepancies are contested by papers written by Ganor and Te'eni [?] and Schröder and Ziefle [?], so this adds a new perspective into this inconclusive, longstanding HCI debate.

Of the literature written on icon accessibility, very few included a Text-Only condition. To some extent, the reason is obvious— in studying iconography, the goal is to condense large amounts of word-based meaning into a visual iota, usually for UI/UX design purposes. But the general results of Mischick Rate suggest that exploration into a middle ground between alienating language entirely and alienating design principles completely.

Of course, these results do not come without room for questioning. One of the main conjectures that will be presented to contextualize the negative impact of IWTL below is that the UI experienced high levels of clutter, to use Grahame et al.'s terminology [?]. Although these results are safely significant, it could be possible that they benefited from lack of clutter, although this would not account for the significant difference between TO and IO specifically. And another possibility is that “digital native” 18-35 year olds might not have been used to associating text with buttons, especially without the explicit hyperlink formatting. Without this prior condition, it might be the case that this TO boost is no longer significant. But with significant results in Mischick Rate that extends equally to both groups, this is unlikely.

4.1.2 Icon with Text Labels Interpretations

The significant results related to the Icon with Text Labels condition were found within models run with Mischick Rate as the dependent variable. In the model of best fit, model 3, IWTL had a coefficient of 6.225 and a p-value of 0.019. Thus, IWTL made accuracy worse (or Mischick Rate higher, specifically by 6.225%) across the board for both Age Groups, as compared to IO and TO. For transparency, these results were not significant in model 4, for either IWTL by itself or the 60+/IWTL interaction term; IWTL was also not significantly worse in Speed for any model.

These results directly conflict with the consensus within the literature. Generally, IWTL conditions reap faster and more accurate results as compared to IO conditions. Dosso and Chevalier observed that labeled icons gave older participants speed boosts

on par with their younger counterparts [?]. This finding is seconded by Wu et al. [?]. Wu et al. specifically found that skeuomorphic icons with text outperformed its flat icon and unlabeled counterparts. Song et al.’s work finds a more dramatic finding, namely that icons presented with text labels enable older users to perform at the same accuracy levels as their younger counterparts [?]. There is some contention from Islam, whose combined quantitative and behavioral study concluded there was no frontrunner between IO and IWTL conditions [?], but the sect within HCI which disagrees with IWTL’s positive performance benefits is small.

So why does IWTL significantly and negatively impact accuracy in this experiment? To propose an explanation, these studies were all conducted in a laboratory setting where icons (or icons with text labels) were shown in isolation. This is not the case in a task-based setting such as this thesis. Icons now exist within a fuller context, which brings along confounding variables such as clutter [?] that do not exist within a controlled, sterile environment. This is both a combination of a change in environment as well as experimental design error, which especially shone through when combined with already complex UIs such as Website 3.

4.2 Limitations

Now, we move towards the limitations of this study, some of which were touched upon by 4.1 already.

The first order of business is acknowledging faulty experimental design. This existed in terms of UI/UX design choices, which was made apparent by the outstanding slowness and error rates on Website 3, and also in terms of general instructions provided. Although much of the error which arose with Website 1 could be attributed with user error, some of it could have originated in unclear instructions. The instruction screen looked something like this:

This was initially altered to provide more clarity (e.g. with wording such as “Task 1: Switch to image search”), but the original version was shown to participants due to a Maze bug.

Another limitation related to experimental design was the decision to host the experiment online and asynchronously through Prolific. There are justifications for this choice (see) and precedent within both HCI and larger fields that feed into HCI, but this could be pointed towards as the main source for the high drop-off rate of roughly 38/9%.

One of the main questions that arose after seeing the skewed proportion of dropped-off older versus younger users (see , e.g. for IO, 6 18-35 users did not finish and 17 60+ users did not finish) was whether “tech-savvy” older users were being disproportionately represented in the dataset, as compared to “non-tech-savvy” older users. Sampling from a pool of already technologically-fluent users was another concern about sampling from Prolific (see), and perhaps the choice to use numbers from only users who fully completed the study exacerbated this.

The reasoning for not including them within the dataset was that they would have contributed to the already high levels of noise within the dataset. That is to



Figure 4.1: An example of an instruction screen provided during the experiment

be expected, but when compounded against the noise that was derived from general user error in relation to the online and asynchronous nature of this study, it becomes concerning. A trend that was qualitatively observed was that users who dropped-off after making it to the UIs would “give up” after realizing they didn’t get the answer immediately. This would result in random clicking, erratic mouse movements, etc., which would overinflate slowness and error within the findings. As mentioned by Mahmud et al., older users specifically exhibit a larger amount of unproductive behaviors stemming from frustration due to obstruction from task-completion as compared to younger Age Groups [?], which might explain the higher drop-off rate for older users as compared to younger.

In general, much of this could have been mitigated by conducting an in-person experiment. Any percentage of the drop-off that was due to technical error would be eradicated, first and foremost. But with user error, in-person experiments can perform “quality control” on participants, which ensure their willingness to give a proper attempt, which despite the monetary reward, is not always the norm on online recruitment platforms such as Prolific. User error also could have been subdued with guidance from experiment leads when confusion arose during the tasks as well.

The last limitation that will be acknowledged is tracking accuracy via Misclick Rate. This is due to its rapid growth rate, which creates poles on both ends and skews reporting statistics such as means and medians. This does not negate its usefulness entirely, especially as a metric which provides insight into an additional dimension of task-completion (accuracy), but it certainly made Results based on these aforementioned statistics less interpretable. It should be mentioned that this statistic is not without backing in the Background Literature [?], but accuracy in general is not reported on as much in part because of an inability to find a more suitable alternative. For instance, converting Misclick Rate into its numerator, the number of incorrect clicks, would still result in two tails, especially with the added phenomenon of random clicking that was prevalent through this online study.

4.3 Extensions

One of the main questions lingering still is whether or not this exciting interaction between TO and 60+ is context-specific. The Websites provided in this study were selected mainly for their common functions in daily life, but without specific criterion for UI design. A future direction that could be pursued is replicating this study with a different set of UIs, or even a more curated sequence of interfaces.

Another main extension would be to replicate this study in-person. Achieving the same sample size would be difficult, but would be a worthy endeavor, especially given the limitations of online studies outlined above. This would also fall more in line with the standards for current HCI experimental design, but these will shift as fields that have adopted online studies more completely into their toolkit continue to influence the world of HCI.

Some other directions that would be less important, but still interesting, to explore would be rotating around the order of appearance of the current UIs (this, alongside

altering the instruction screens, could remove the outlier nature of Website 1), and changing what type of icons are displayed (currently, icons more akin to flat icons discussed in are used, but it could be worth exploring skeuomorphic iconography or alternative forms of iconography instead.

4.4 Conclusion

As technology continues to evolve in rapid, perhaps even unsustainable, rates, the needs of users are required to be centered more than ever before. User-centric design is a real concept and is touted as the golden standard of product development in the technological sphere, but is not all-inclusive. Instead, software development and UI/UX design center the conventional user, someone with fluency in technological navigation and no lack in cognitive, visual, motor, etc. ability. This myth of the everyday user sidelines many people of varying abilities, including user populations such as the elderly who experience unique combination of disabilities associated with aging.

Age-based accessibility research in Human-Computer Interactions aims to unearth various ways of making the technological experiences of older users easier and more productive. These findings require flexibility, as age-based declines do not have a definitive prototype.

Icon Type is one of the research fields within this discipline which has been hotly contested. Results on whether certain alterations benefit users, such as skeuomorphic versus flat icon design and with or without text labels, are all in flux. It is also a field that has primarily existed within the context of laboratory experimentation. But with the need for more conclusions, especially conclusions that are relevant to the everyday use cases of these icons, there became a gap that this thesis could fill.

This thesis explored whether different Icon Types, namely Icon-Only, Text-Only, and Icon with Text Labels, when presented in the context of a web-based task, had different age-based performance effects. Each participant was assigned a Icon Type condition and solved the same sequence of 6 UIs online and asynchronously.

A total of 240 users completed this experiment, with roughly 80 users per Icon Type and 40 users per Icon Type and Age Group subdivisions. After excluding Websites 1 and 3, the main finding from this study was that the Text-Only condition disproportionately positively impacted the Speed of the 60+ Age Group. Alongside that, Text-Only was proven to be significantly positive for accuracy, and Icon with Text Labels was proven to be significantly negative for accuracy.

Thank you.

Appendix A

The First Appendix

Appendix B

The Second Appendix, for Fun

